

Washington Crime Analysis Project

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2024-01-23

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1 Introduction

Three large cities in Washington - Bellevue, Spokane, and Tacoma - make regular crime reports available. Spokane releases its data on a weekly basis. This project collects and parses that data, and provides up-to-date analysis. The same data is formatted and submitted annually to the FBI NIBRS program as well. A lot of work is required to prepare the data for submission to NIBRS, but it's more granular and includes detailed information that weekly reports lack. Unfortunately, it takes a year or more for current data to be available in the NIBRS portal. It will most likely be late-2022 or

early-2023 before 2021's data is ready. We use the weekly reports to identify trends happening now and provide up-to-date analysis. NIBRS-based analysis is a separate project. The additional detail from years past informs our weekly analysis and helps us frame questions better.

When this project began it was strictly focused on crime in Spokane. It's since expanded to include other cities with similar population size that also make crime data available regularly. Here are the three covered cities and their population based on Census Bureau estimates for 2020:

City	Estimated 2020 Population
Spokane	222,050
Tacoma	219,995
Bellevue	148,073

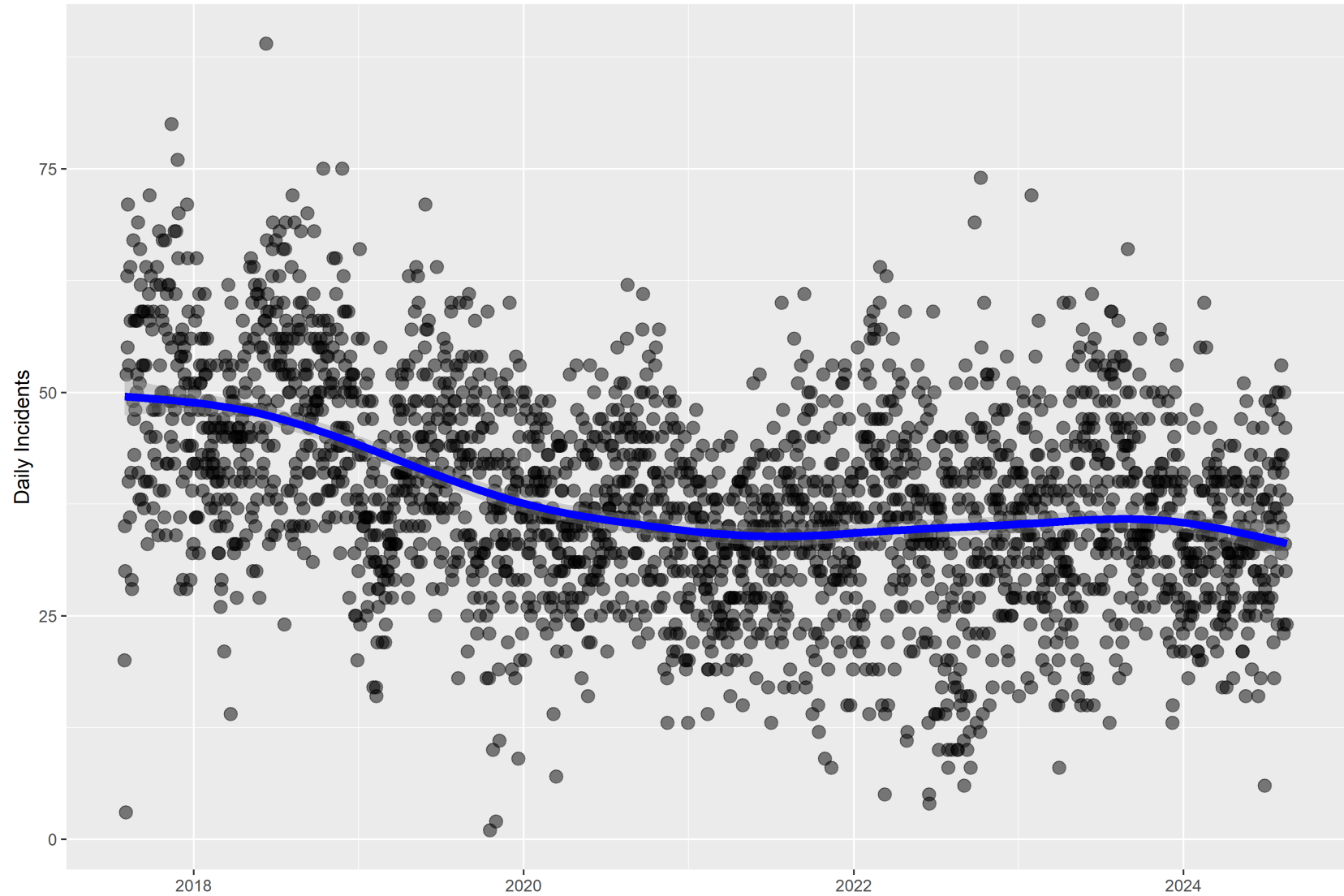
I'd have liked to include the city of Vancouver, as it has a population between Bellevue and Tacoma, but they don't make regular crime data reports available. Seattle is so much larger than all the other cities that I've excluded its crime analysis from this project.

2 Incidents Over Time

We visualize incidents over time by totaling up incidents and plotting them daily. Blue and red lines show average values using generalized additive model (GAM) and linear model (LM), respectively. Then we break down data for time series analysis and identify seasonality and trends.

2.1 Spokane

Spokane Total Incidents

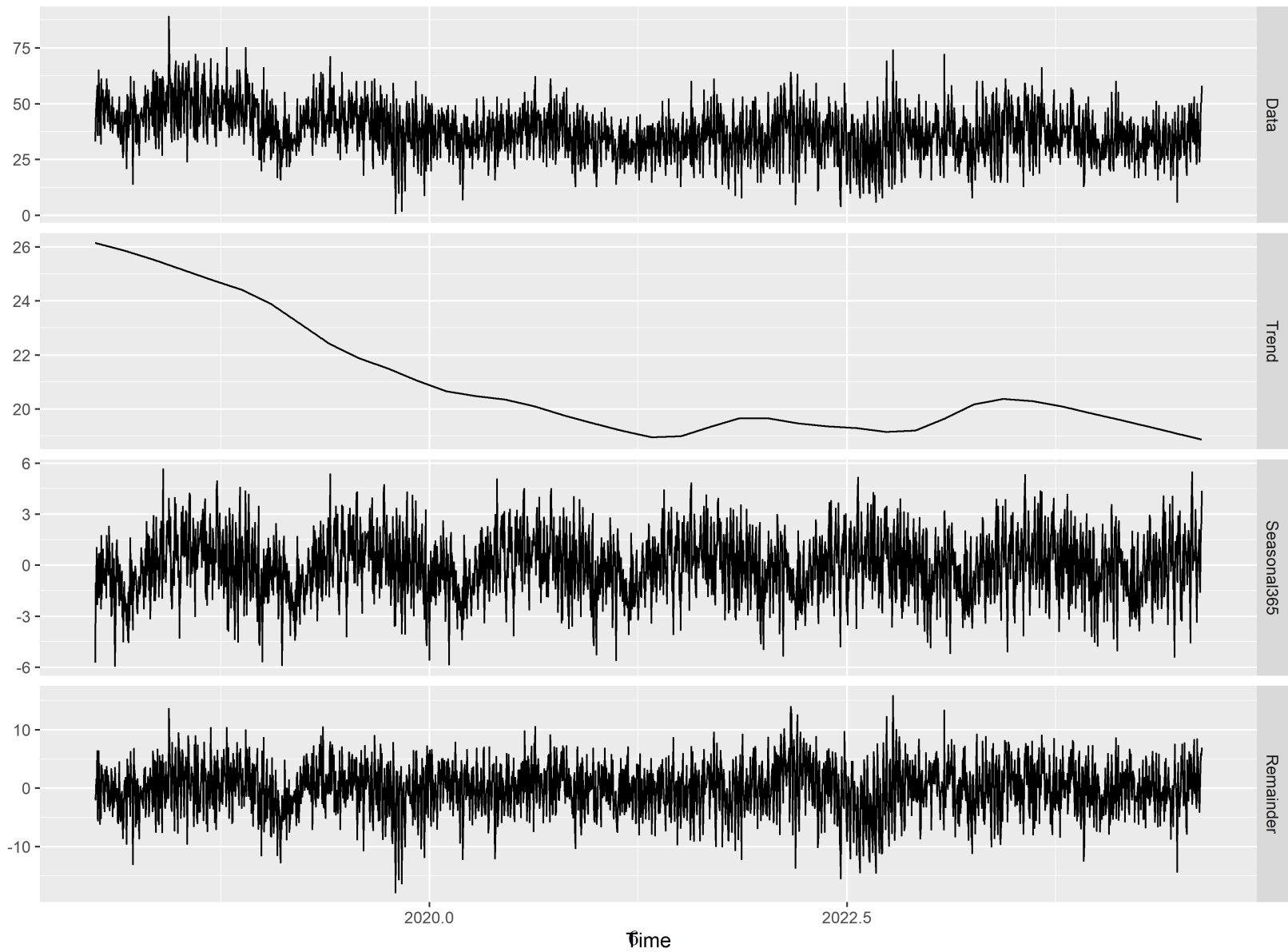


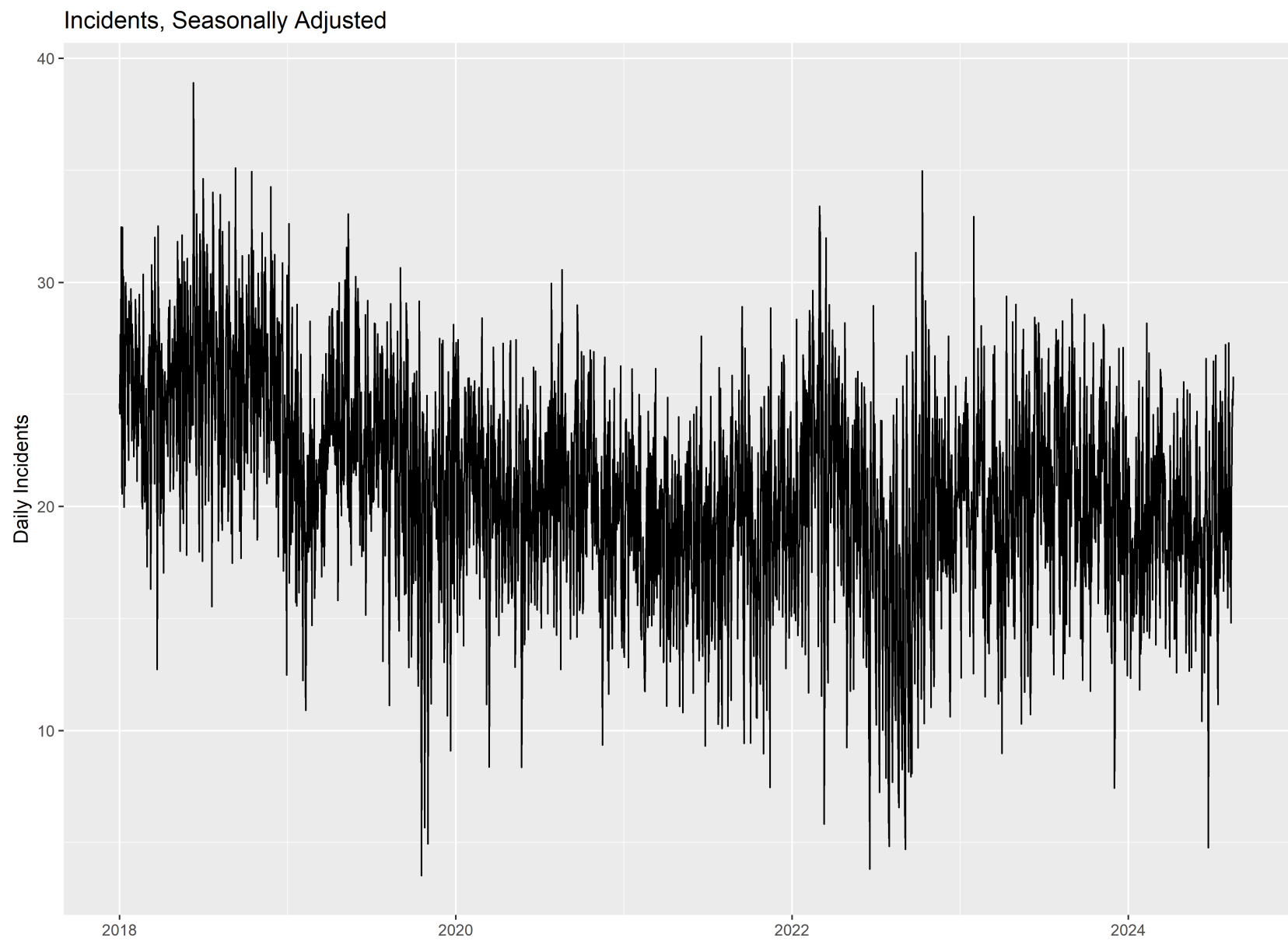
Created 2024-08-25; Data Source: Spokane PD CompStat
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2.2 Tacoma

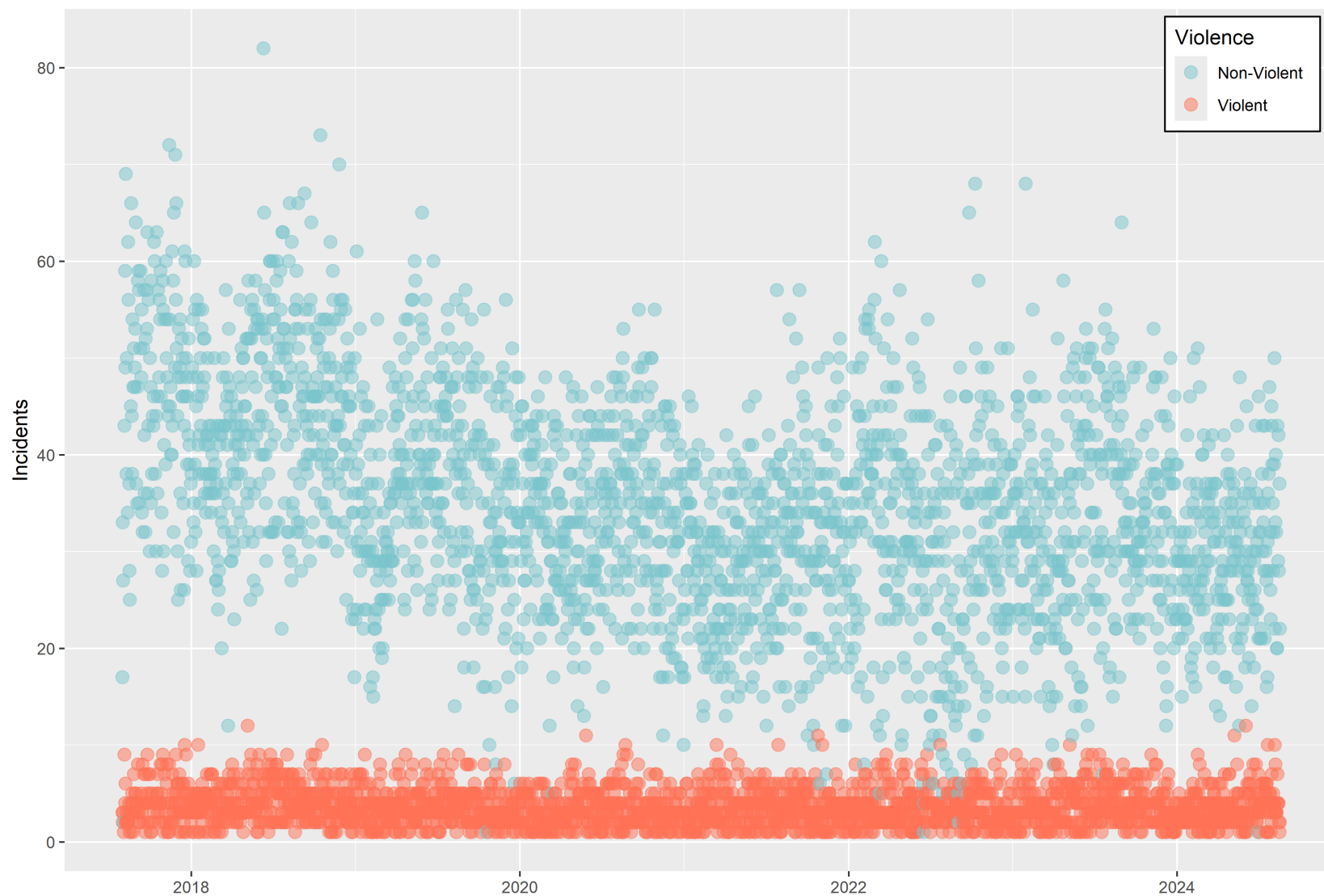
2.3 Bellevue

Spokane Crime Seasonal Decomposition



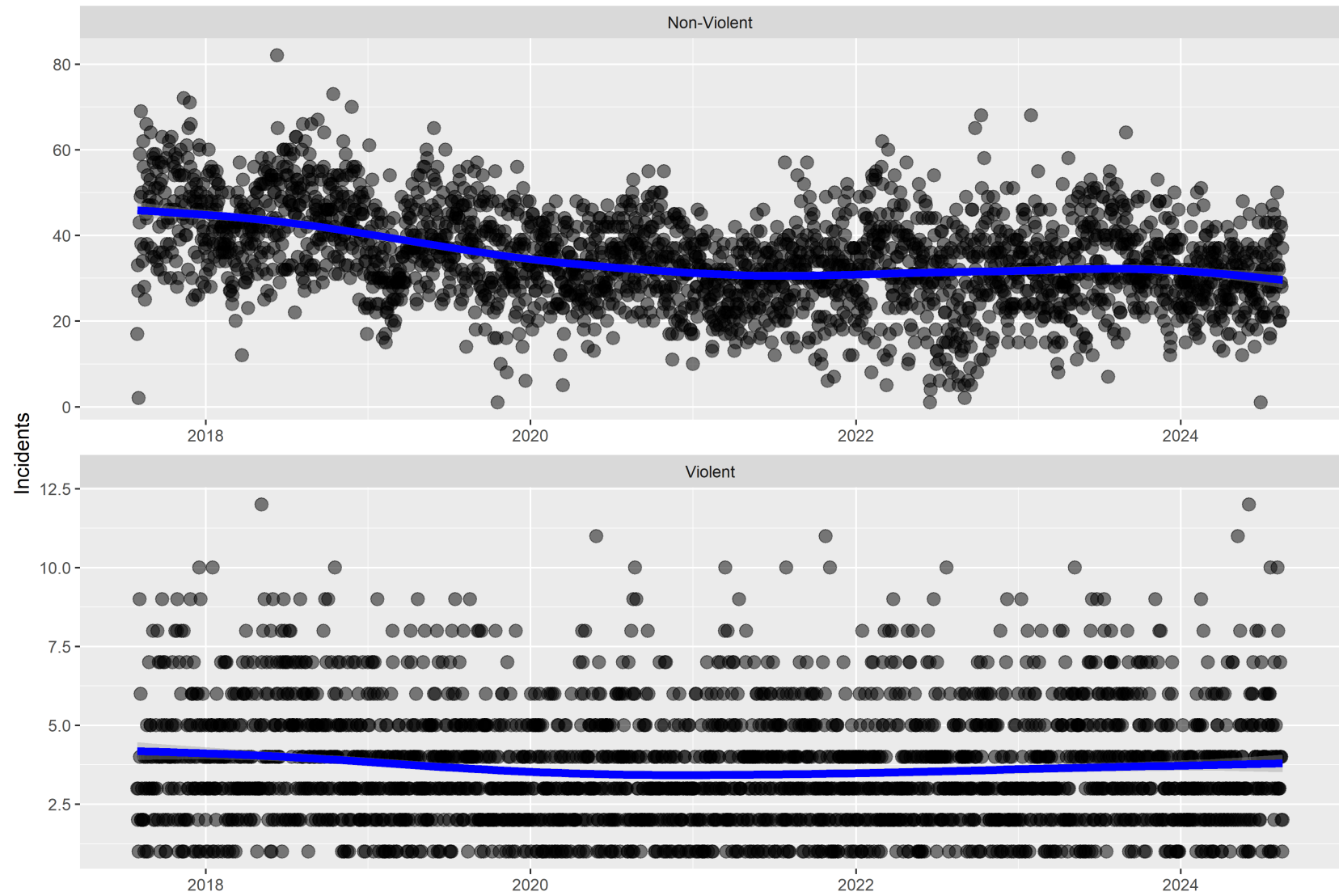


Daily Non-Violent & Violent Incidents

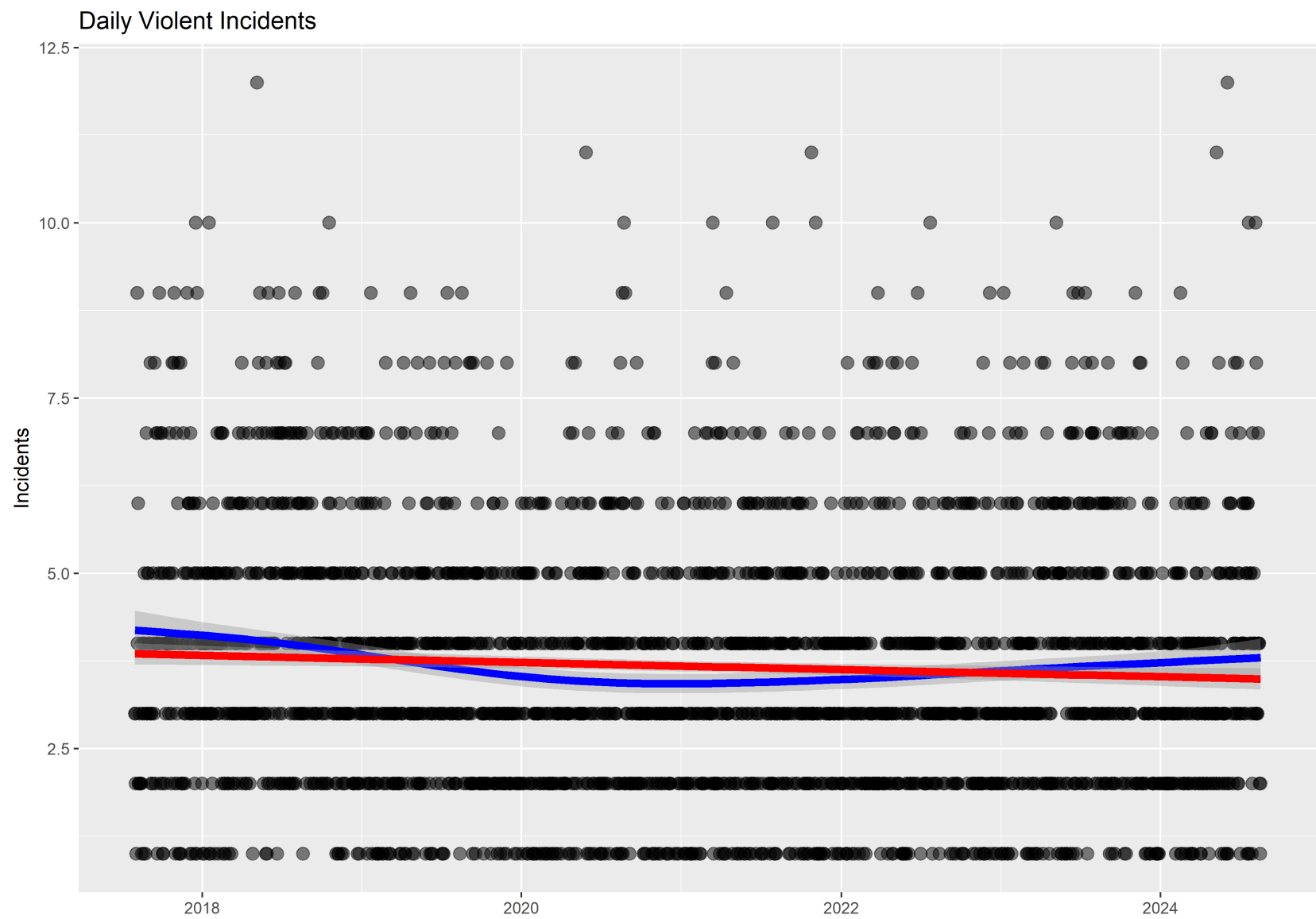


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Daily Non-Violent & Violent Incidents

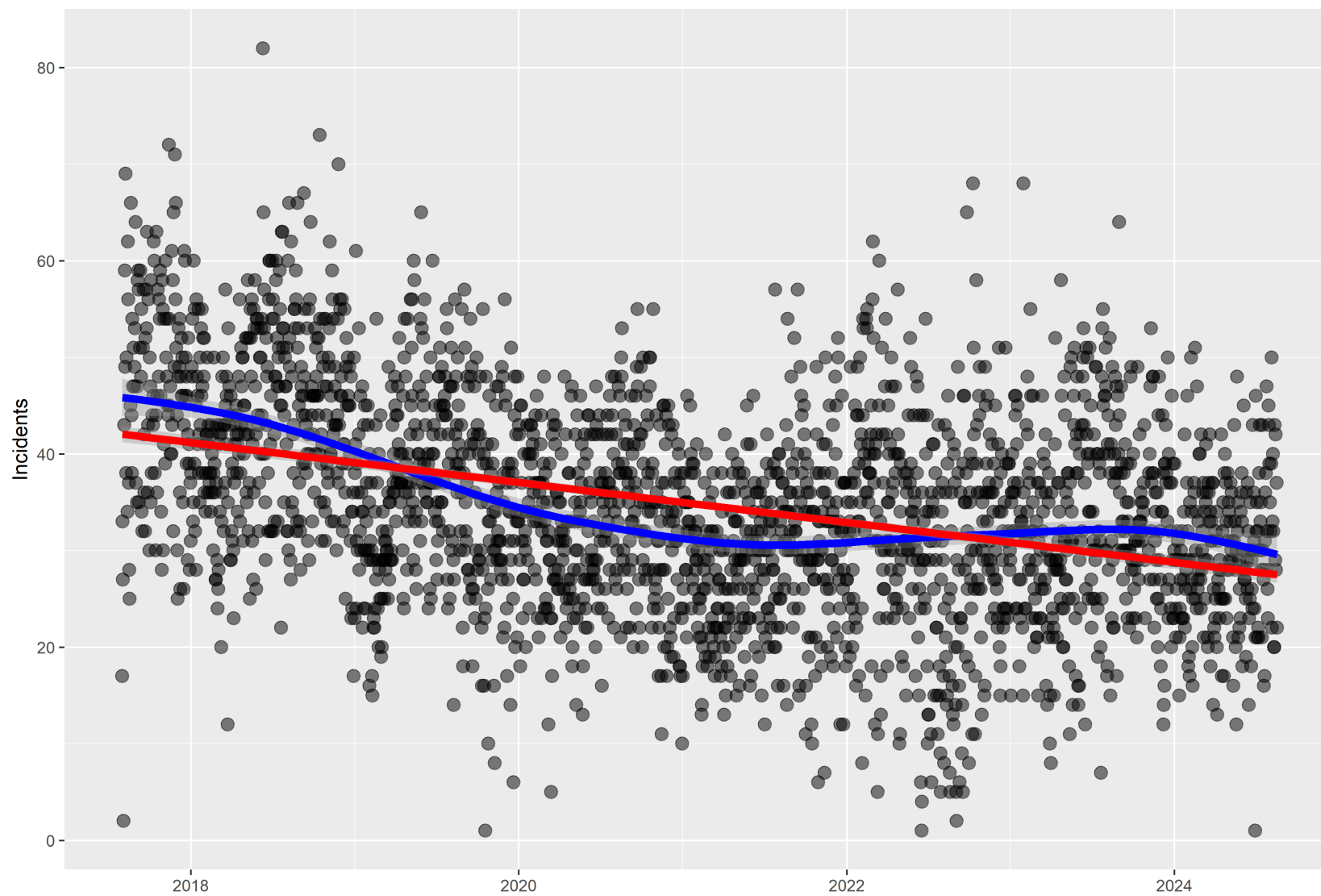


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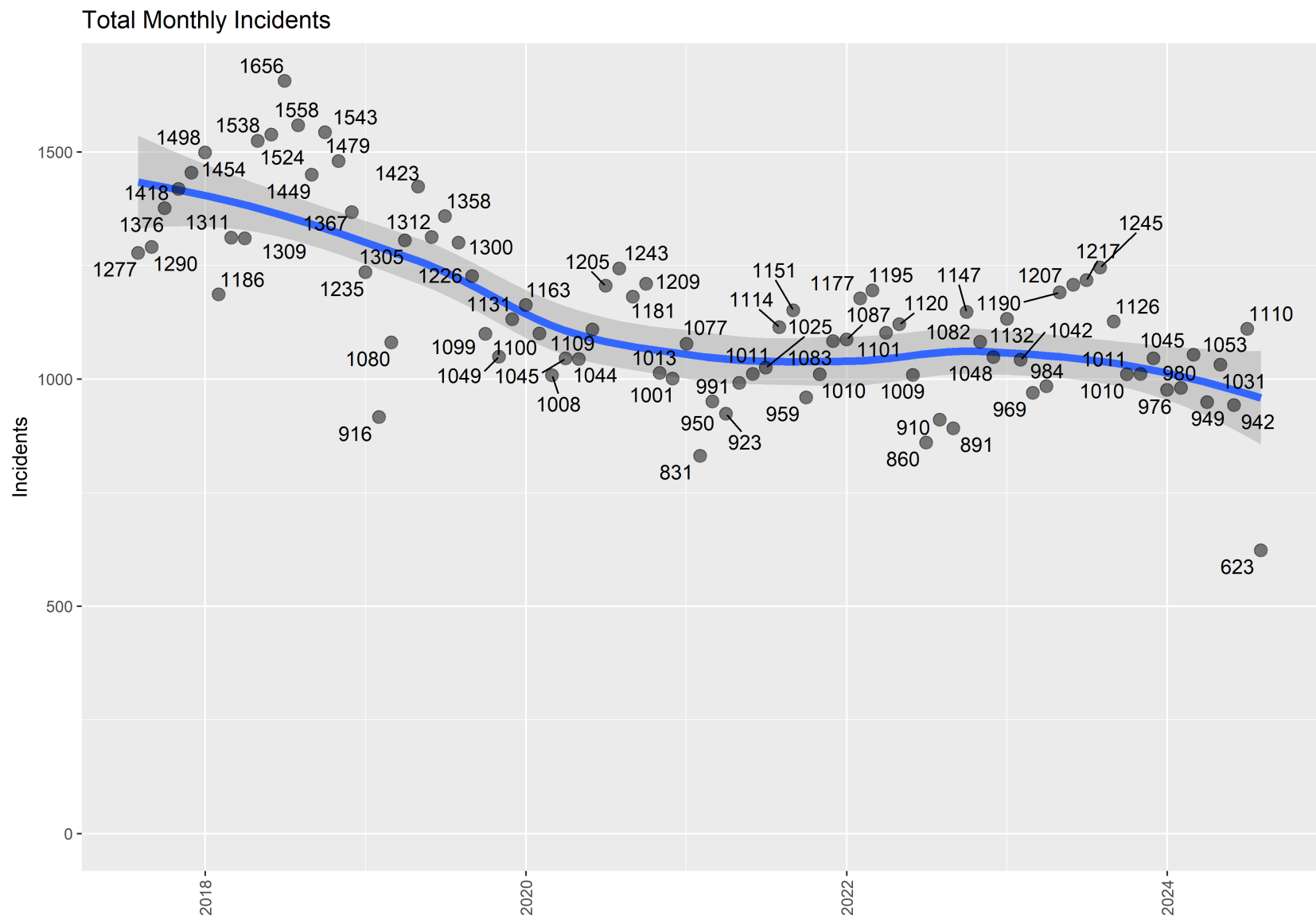


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Daily Non-Violent Incidents

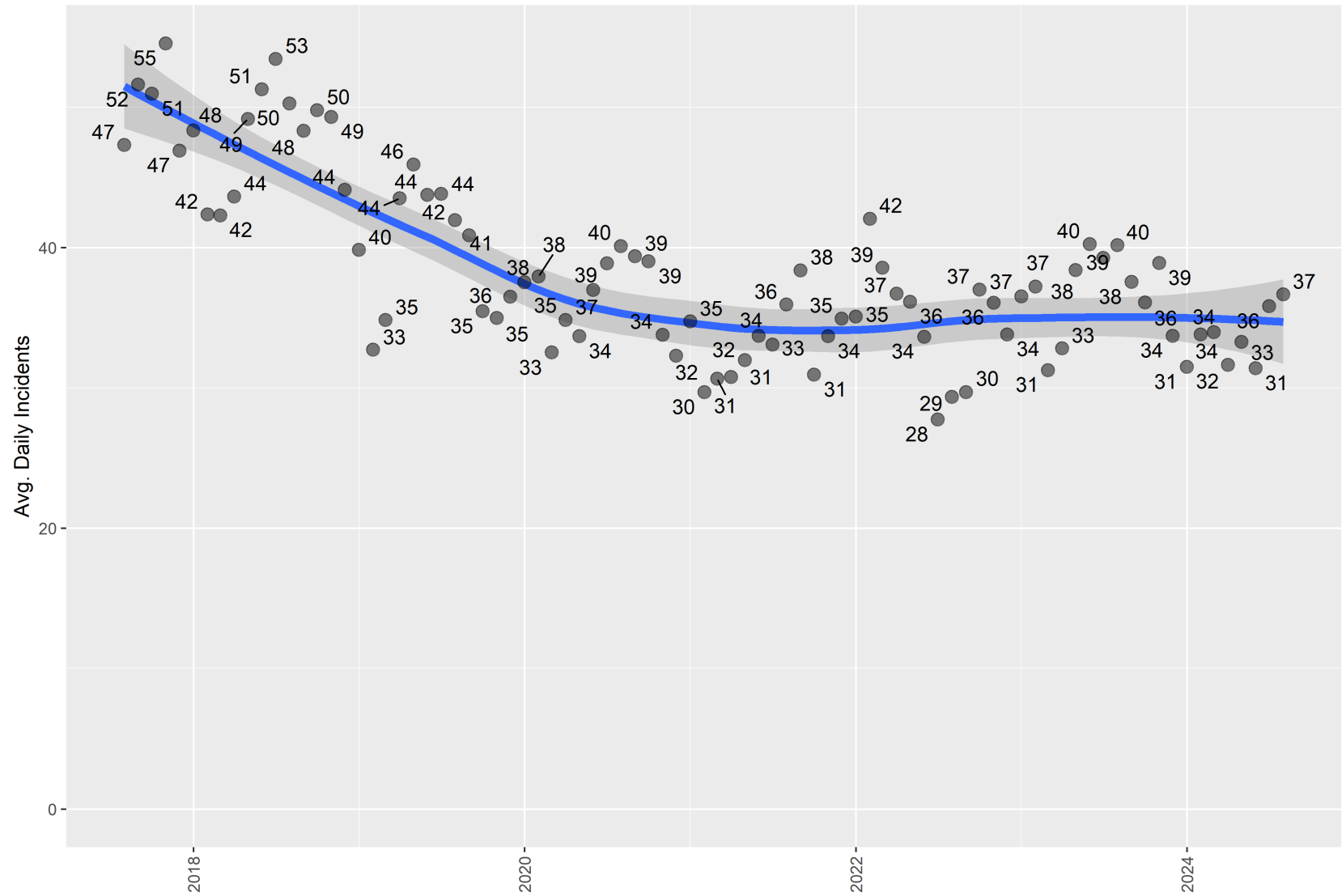


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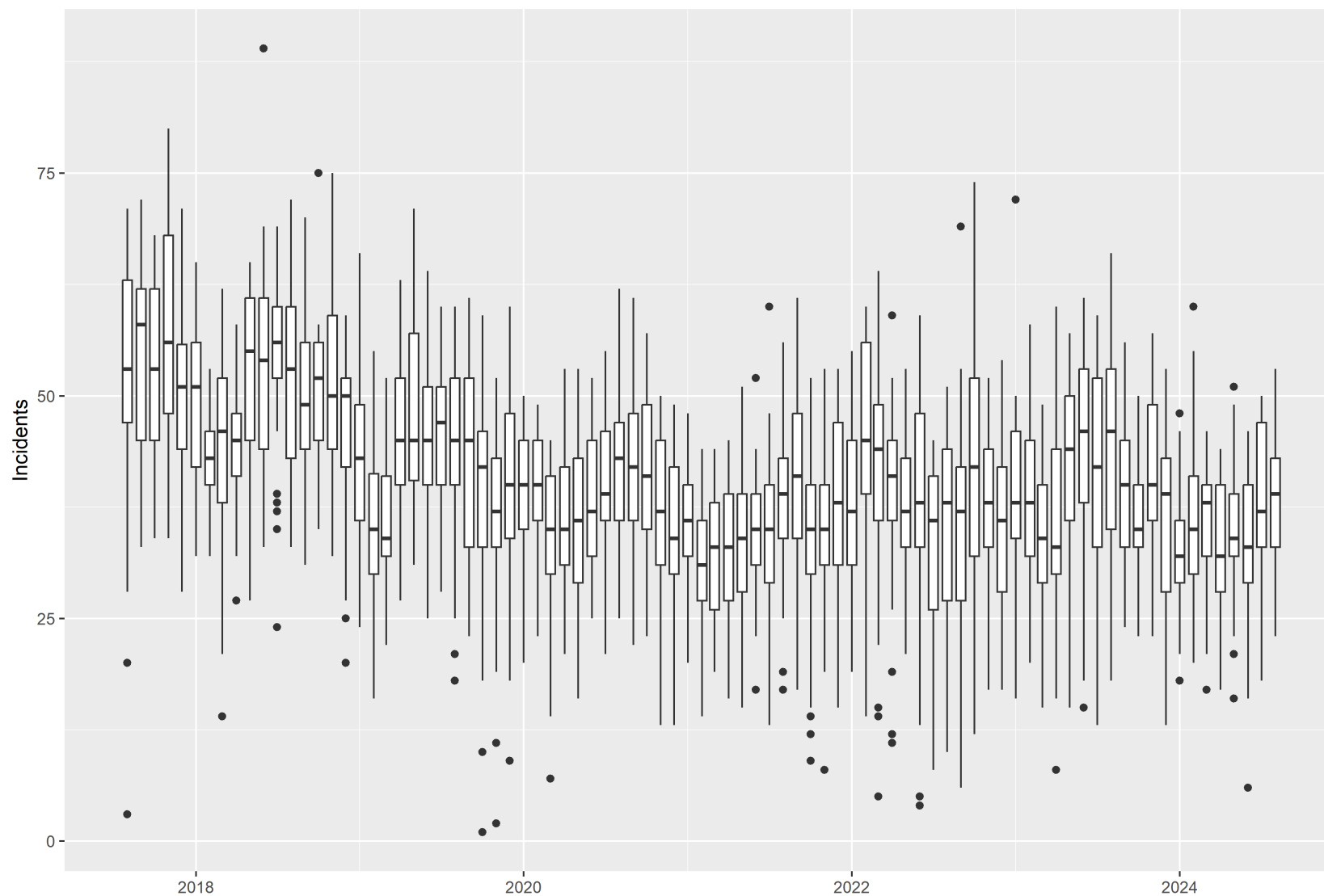
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Average Daily Incidents by Month

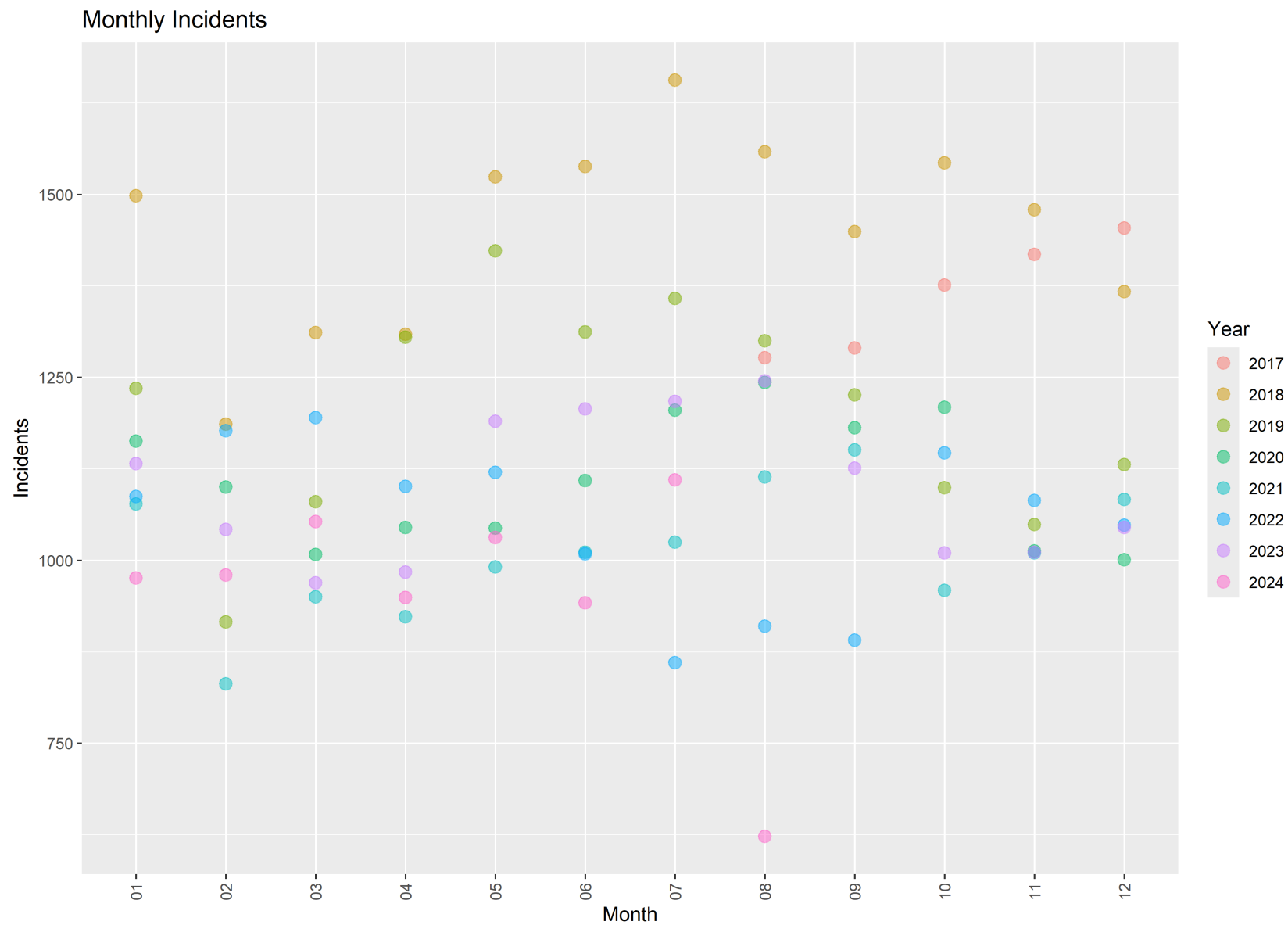


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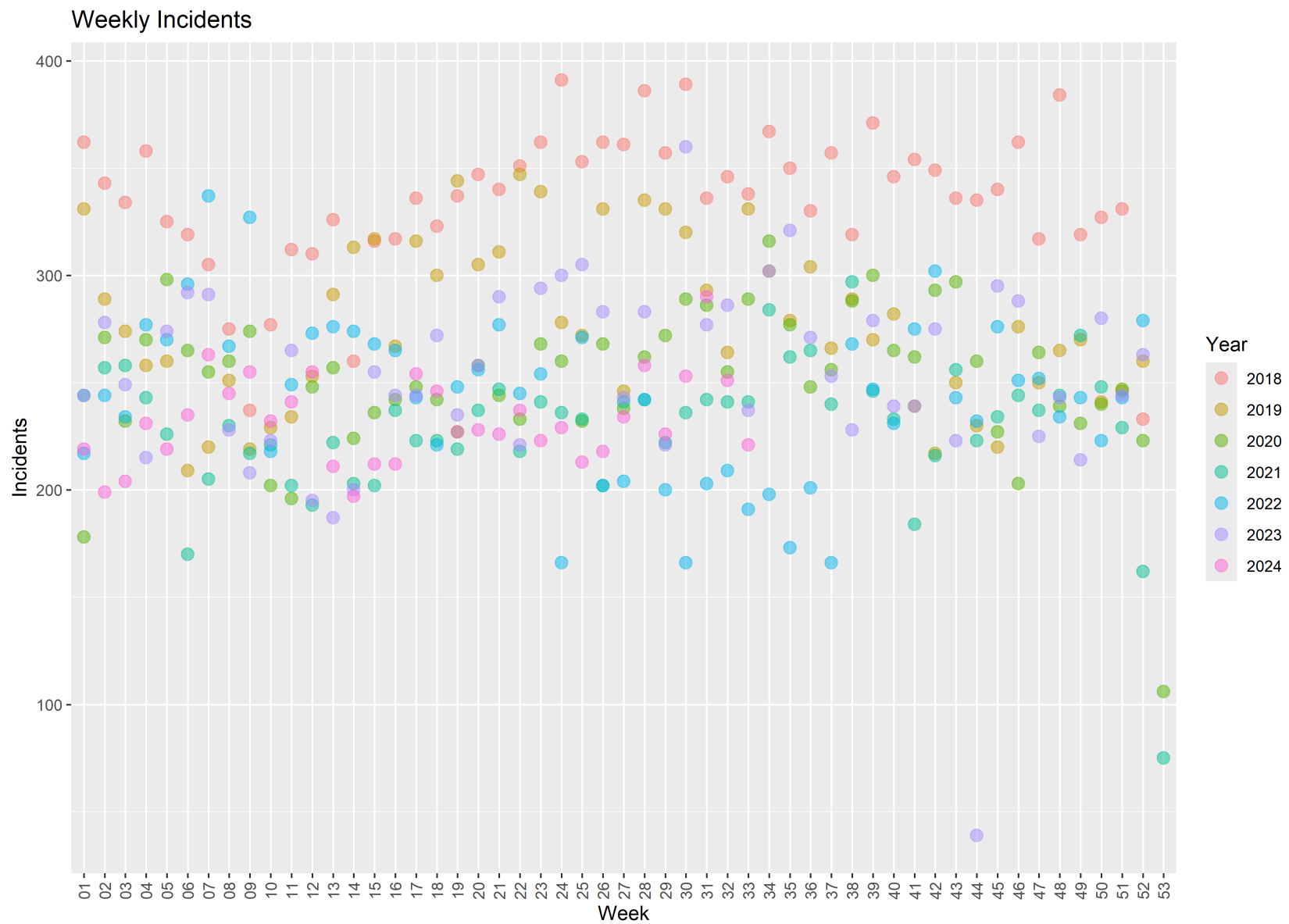
Daily Incidents by Month



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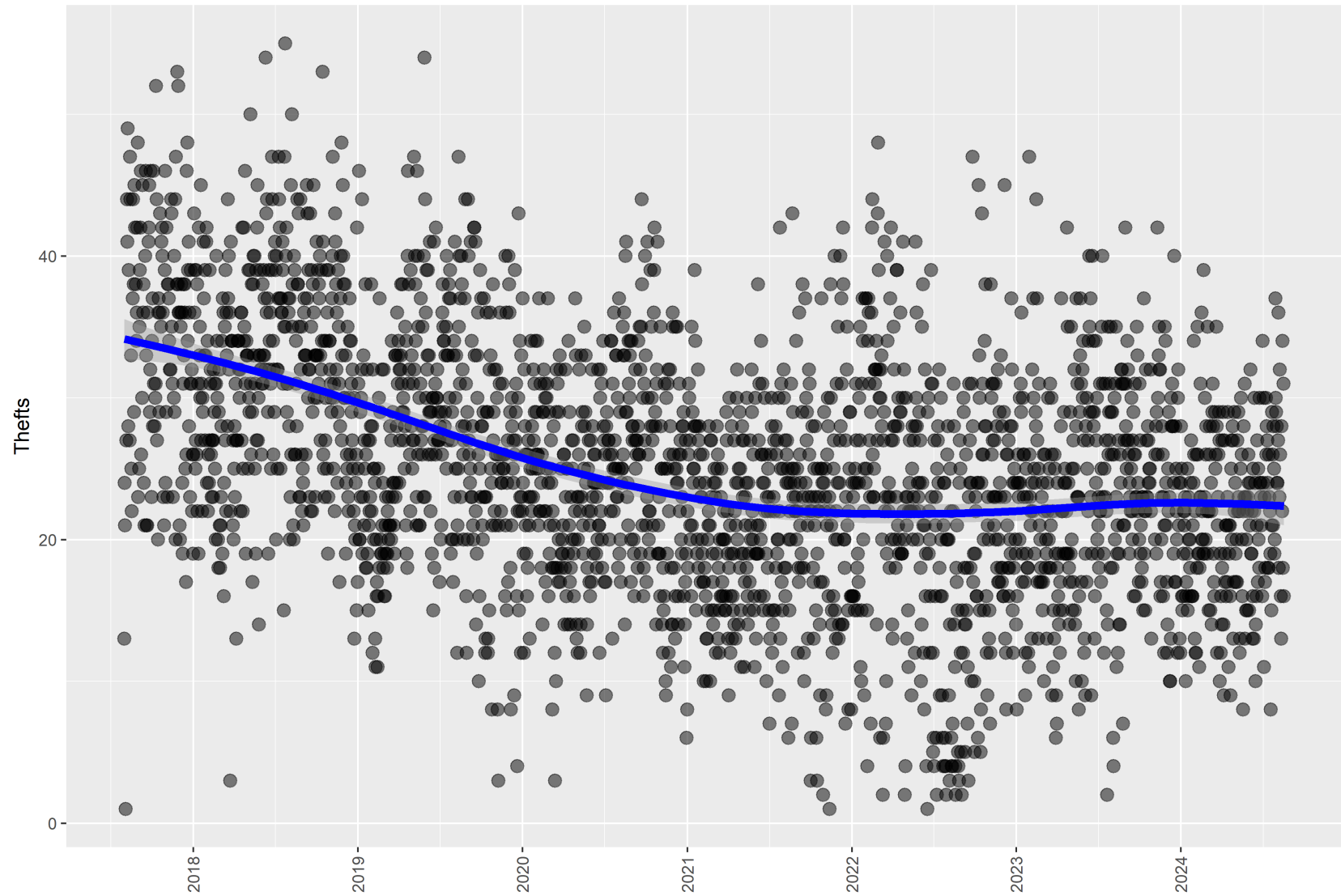


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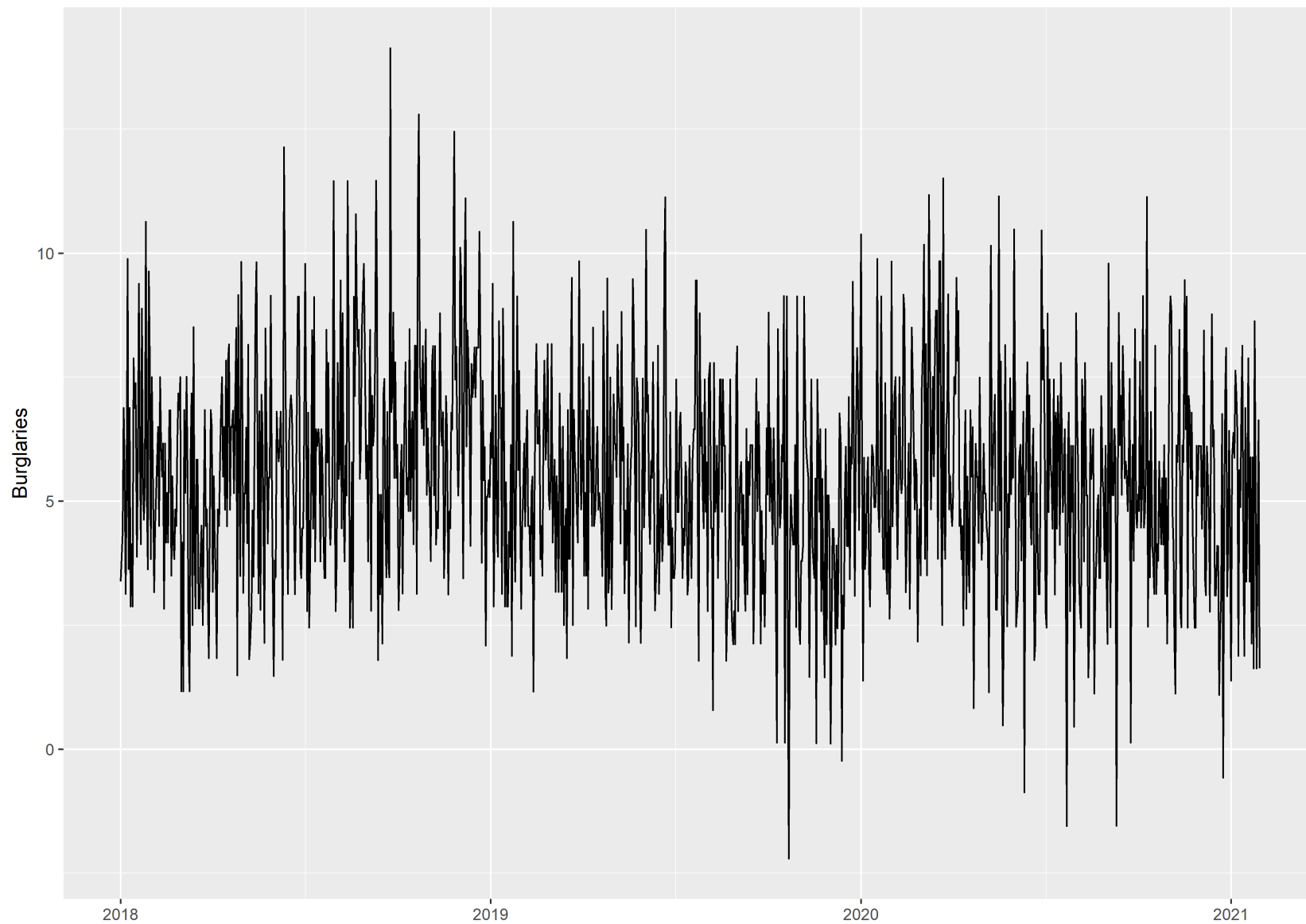
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Overall Thefts

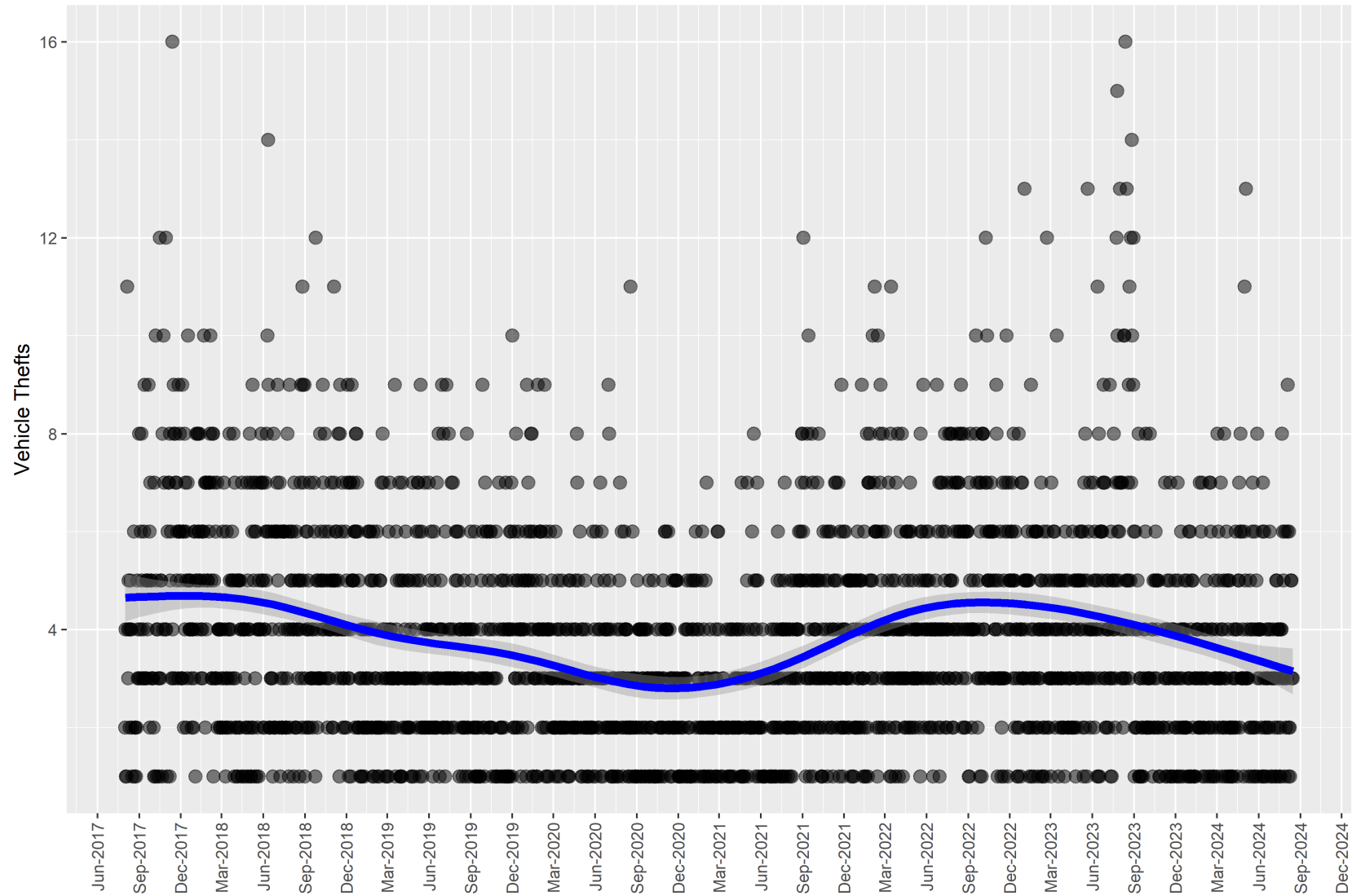


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Burglaries, Seasonally Adjusted

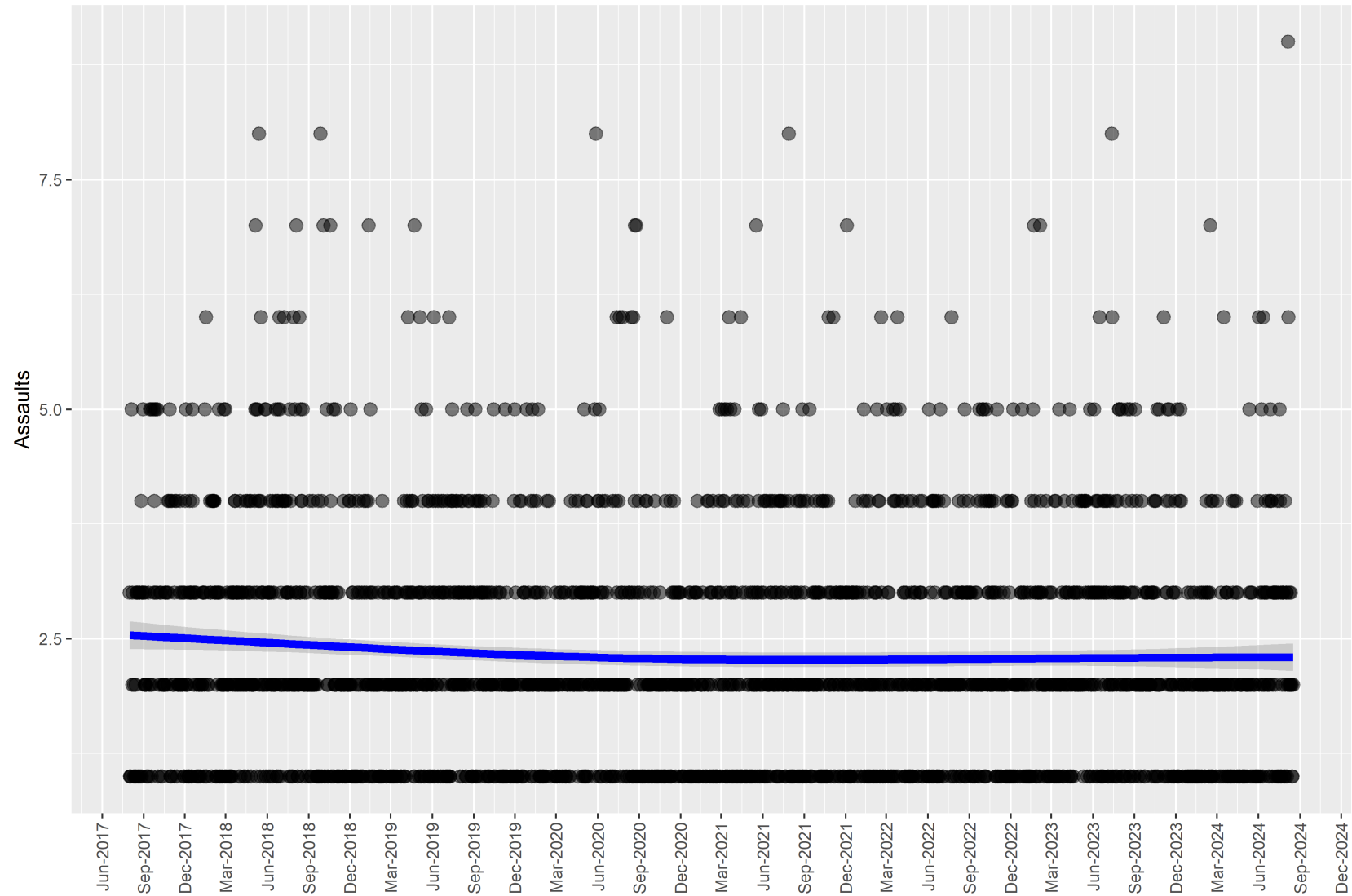


Vehicle Thefts



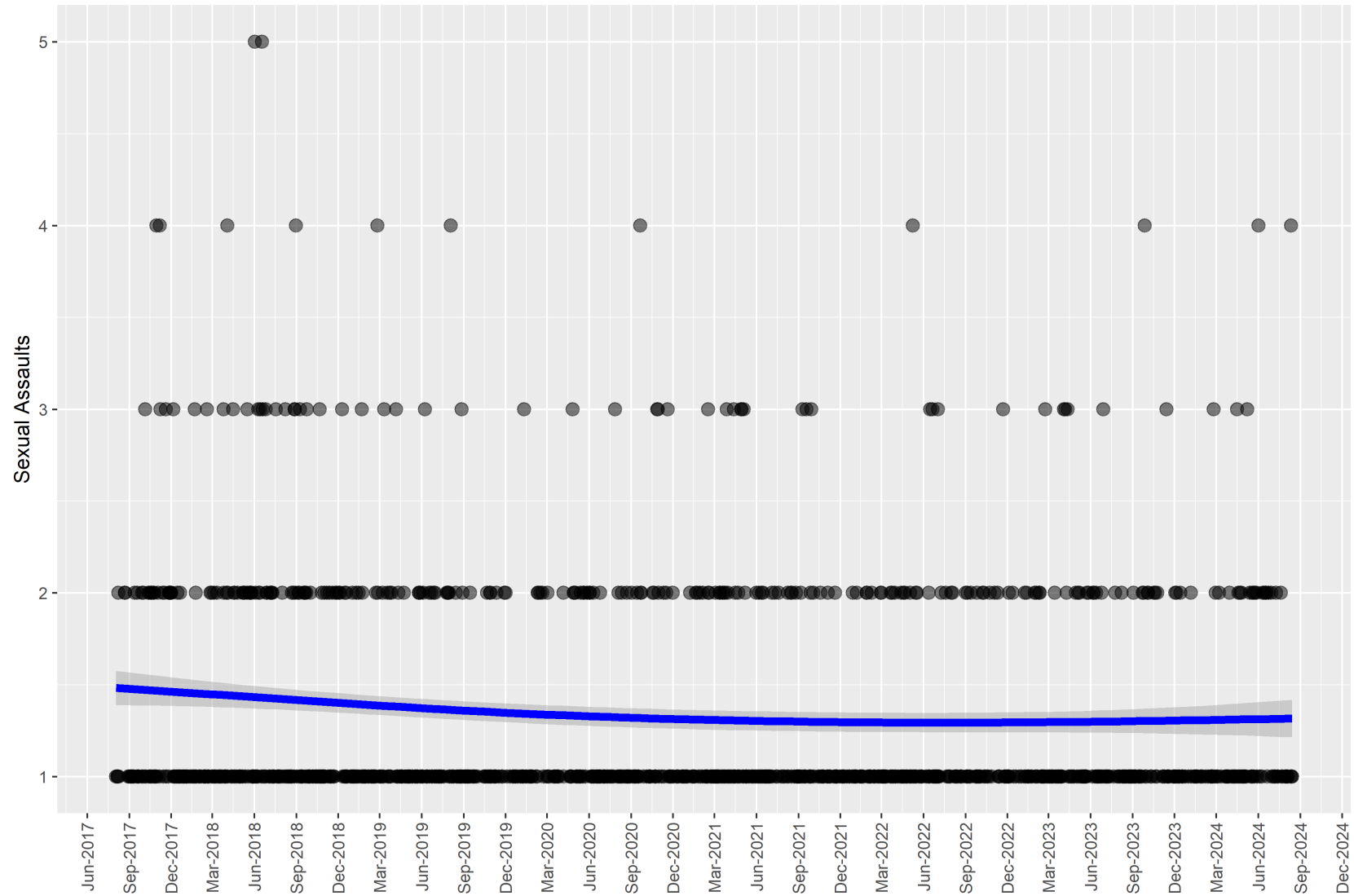
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Assaults



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Sexual Assaults

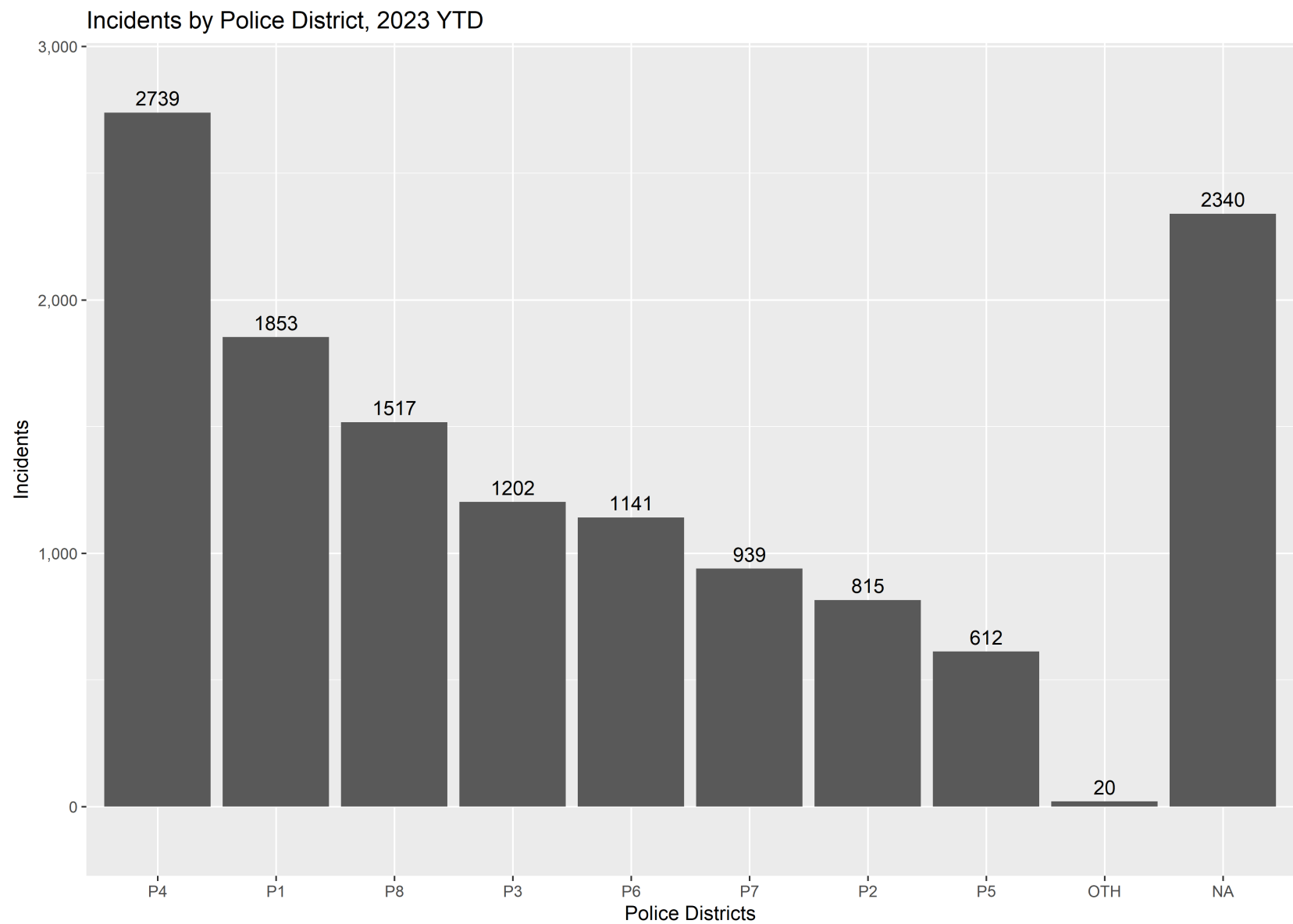


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3 Police Districts

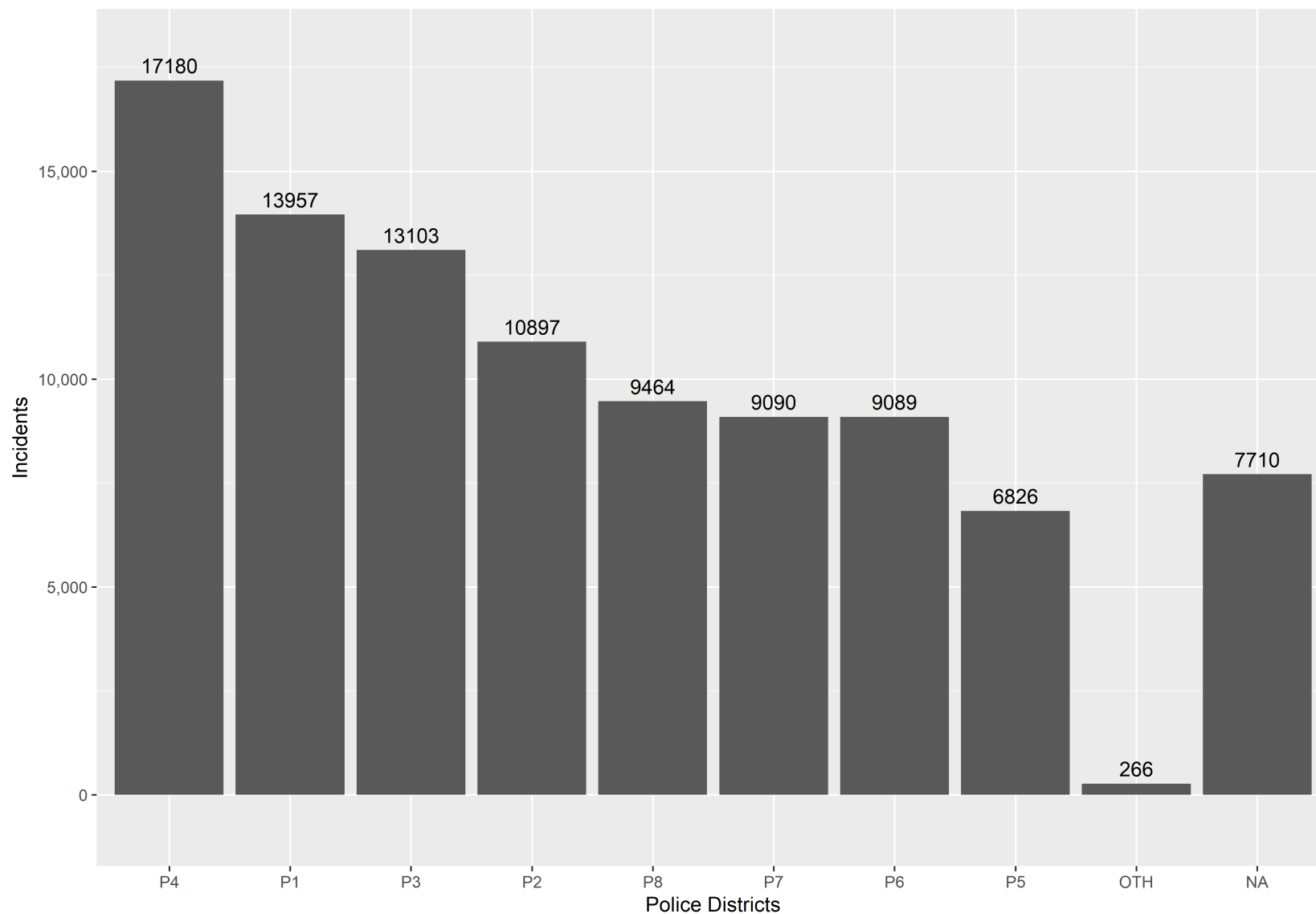
Spokane has a total of eight police districts. Those eight are divided evenly between two main groups, the North and South Police Service Areas. The Spokane River separates the two service areas, with Division St. and other major thoroughfares serving as boundaries. Districts P1 - P4 are north of the Spokane River. Districts P5 - P8 are south of the river. Some incidents in the CompStat reports are listed as “OTH”, meaning “Other”.

- P1: Northwest District
- P2: Central District
- P3: Neva-Wood District
- P4: Northeast District
- P5: South Central District
- P6: Southeast District
- P7: Southwest District
- P8: Downtown District
- OTH: Other District



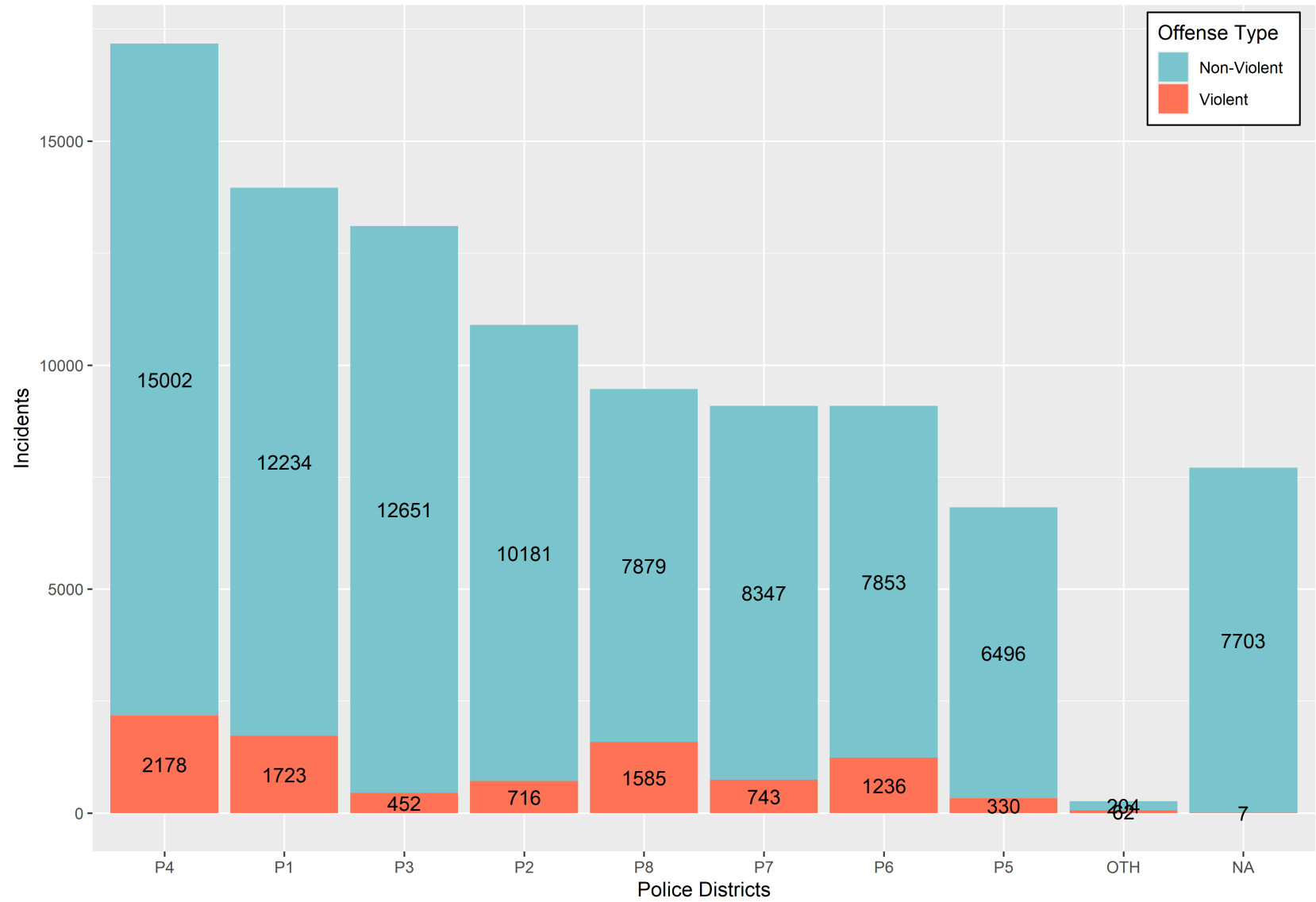
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Incidents by Police District, 2017 - 2023



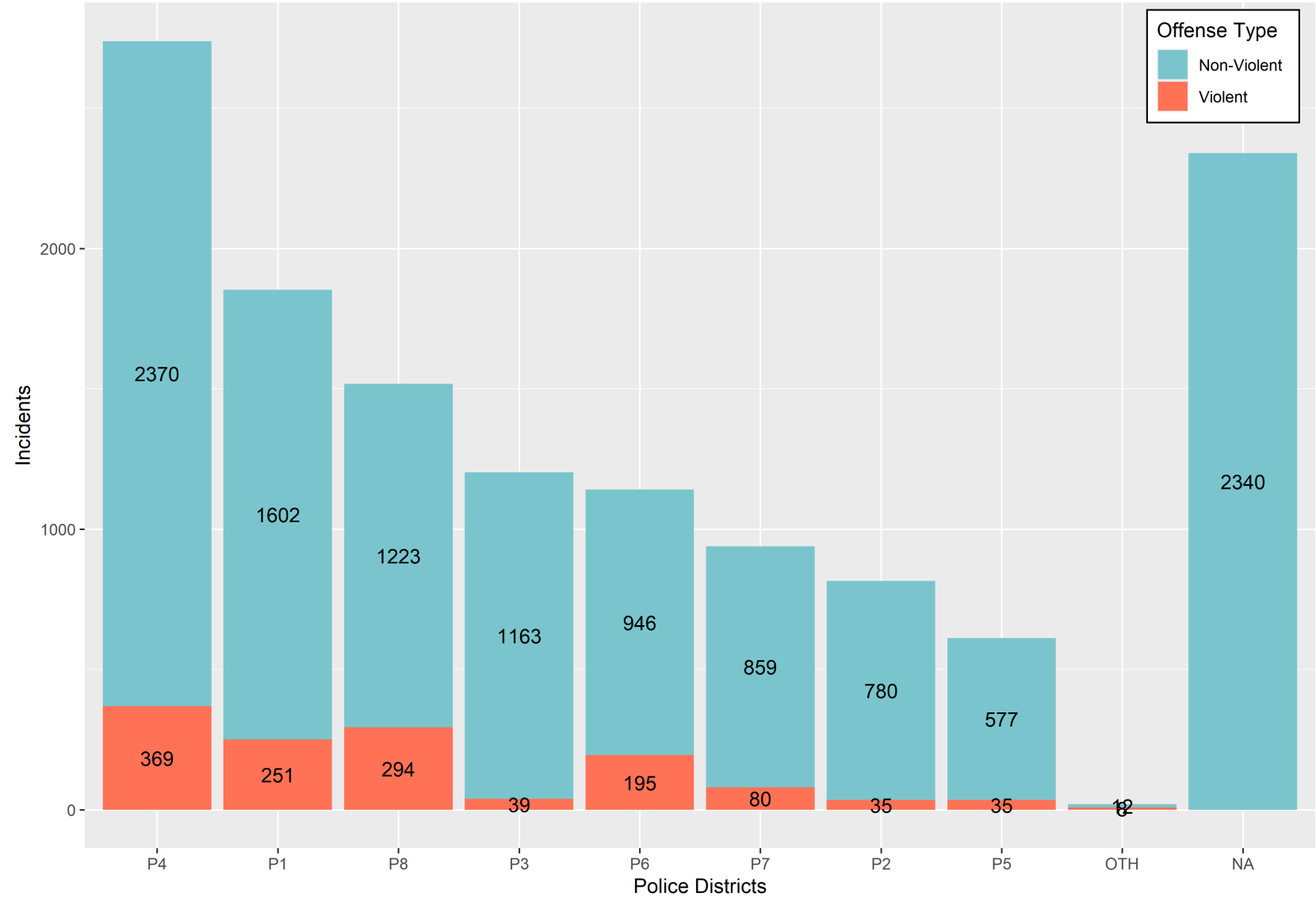
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Incident Violence by District, 2017-2023



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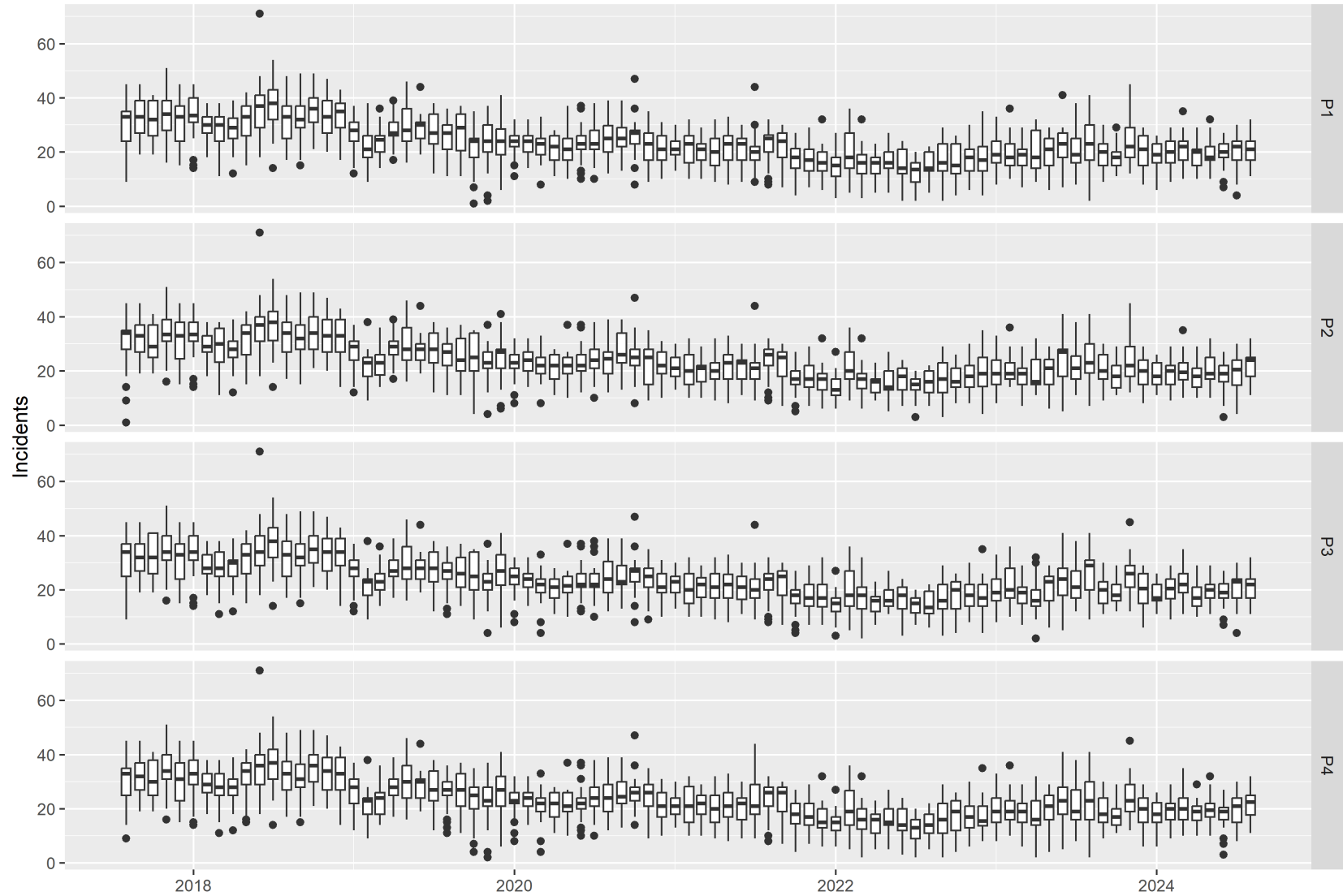
Incident Violence by District, 2023



Created 2024-08-25; Data Source: Spokane PD CompStat
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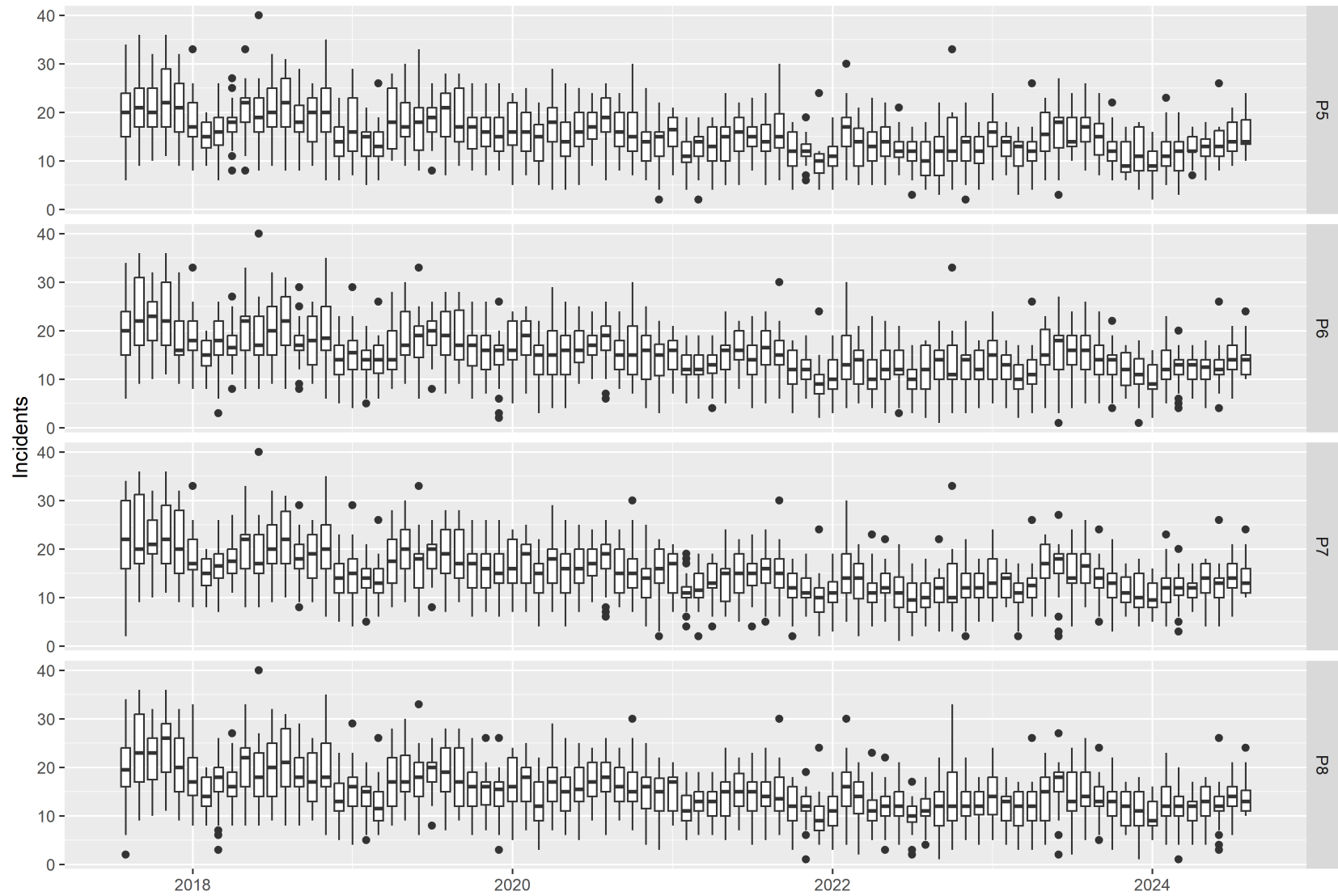
3.1 North, South Districts

North Policing Districts, Daily Incidents by Month



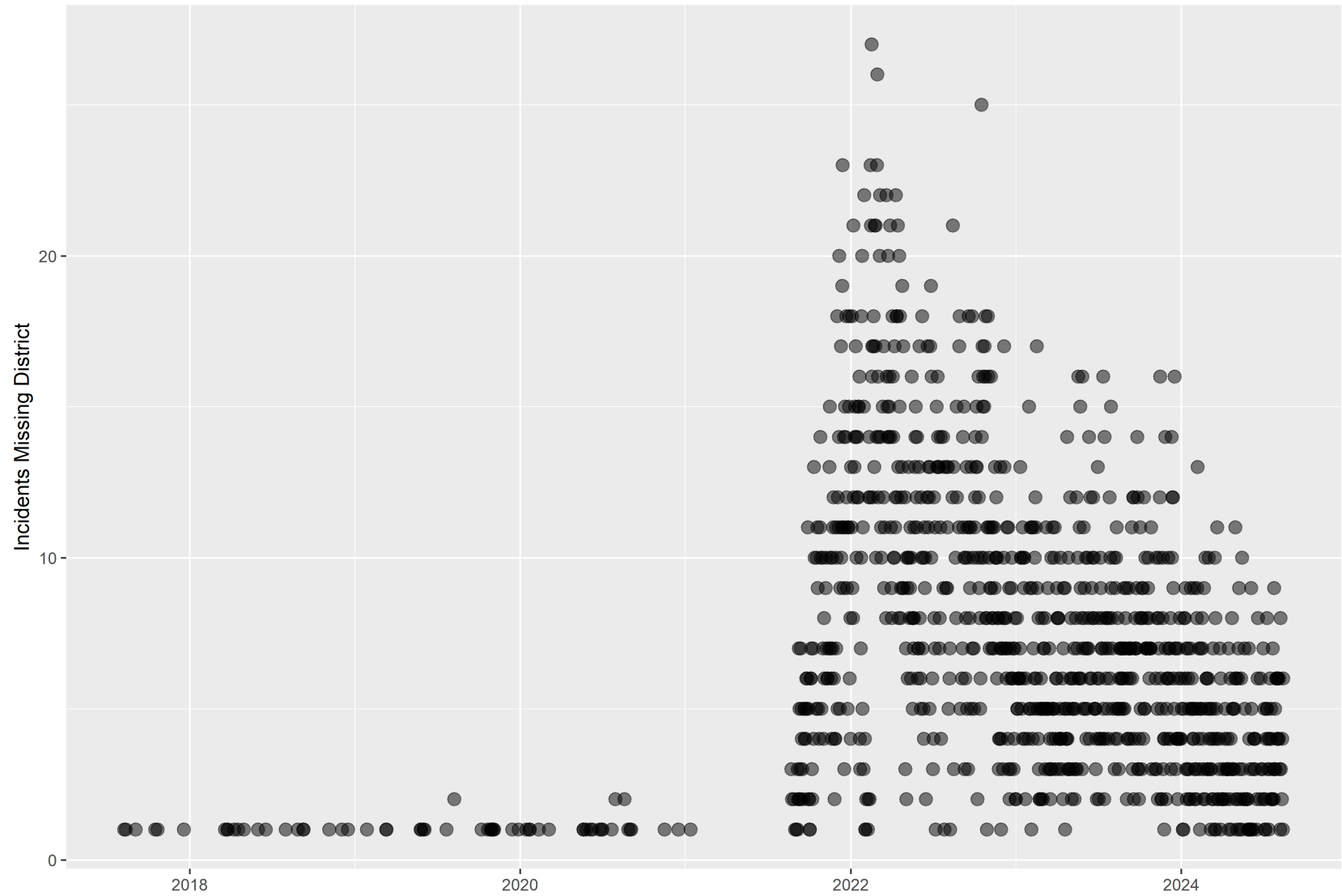
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South Policing Districts, Daily Incidents by Month



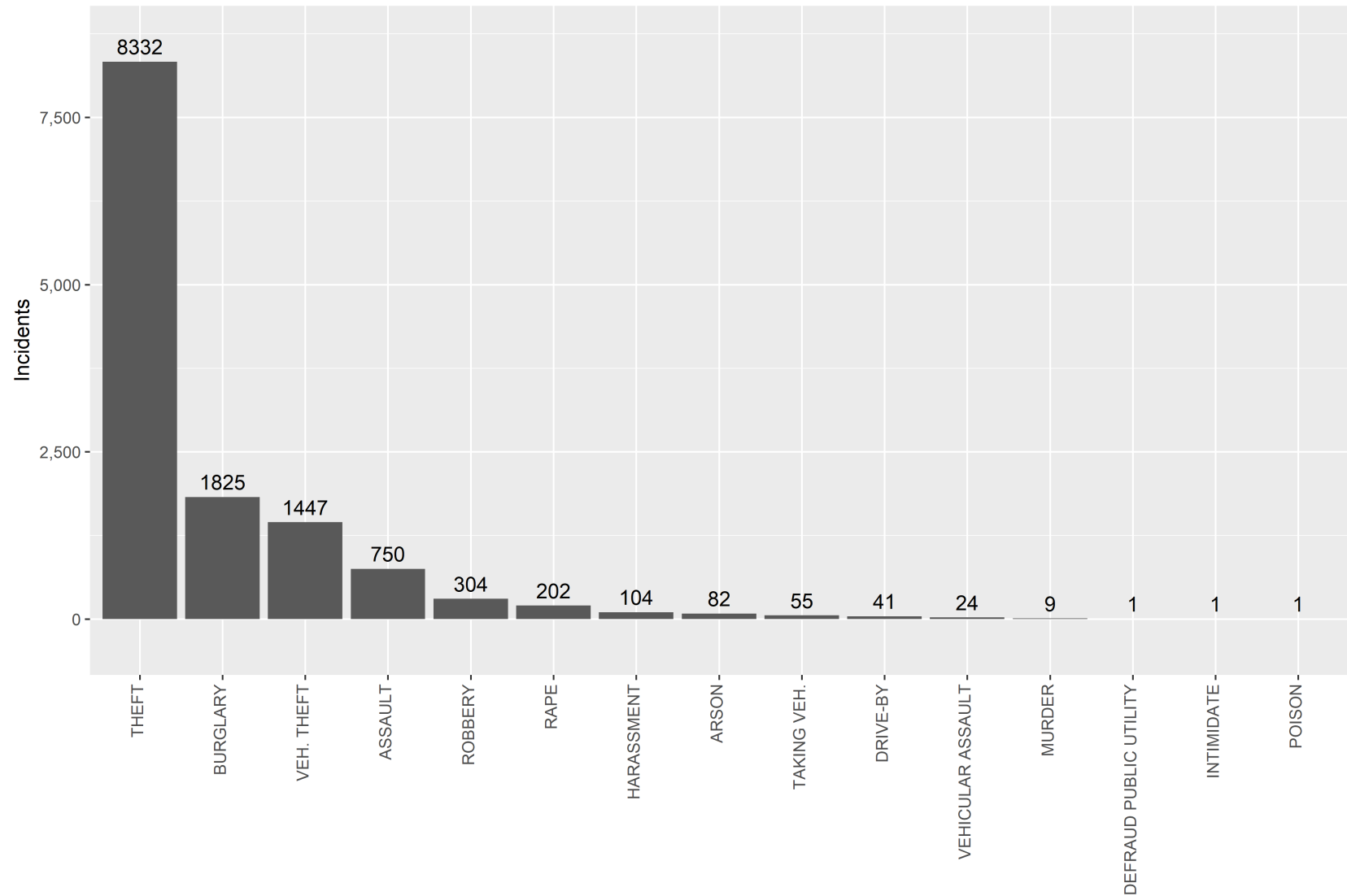
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Incidents Without a Police District



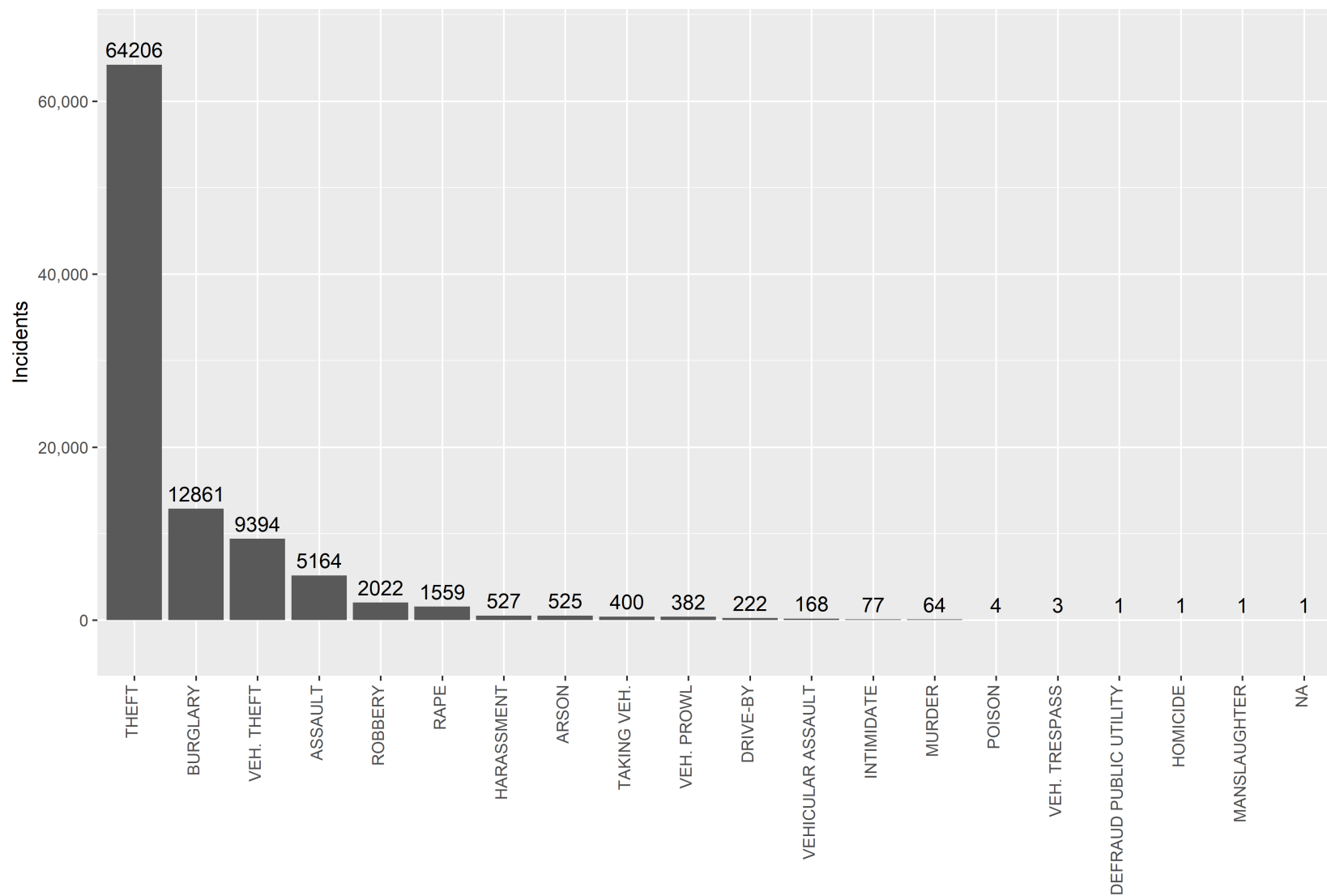
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Incident Types, 2023 YTD



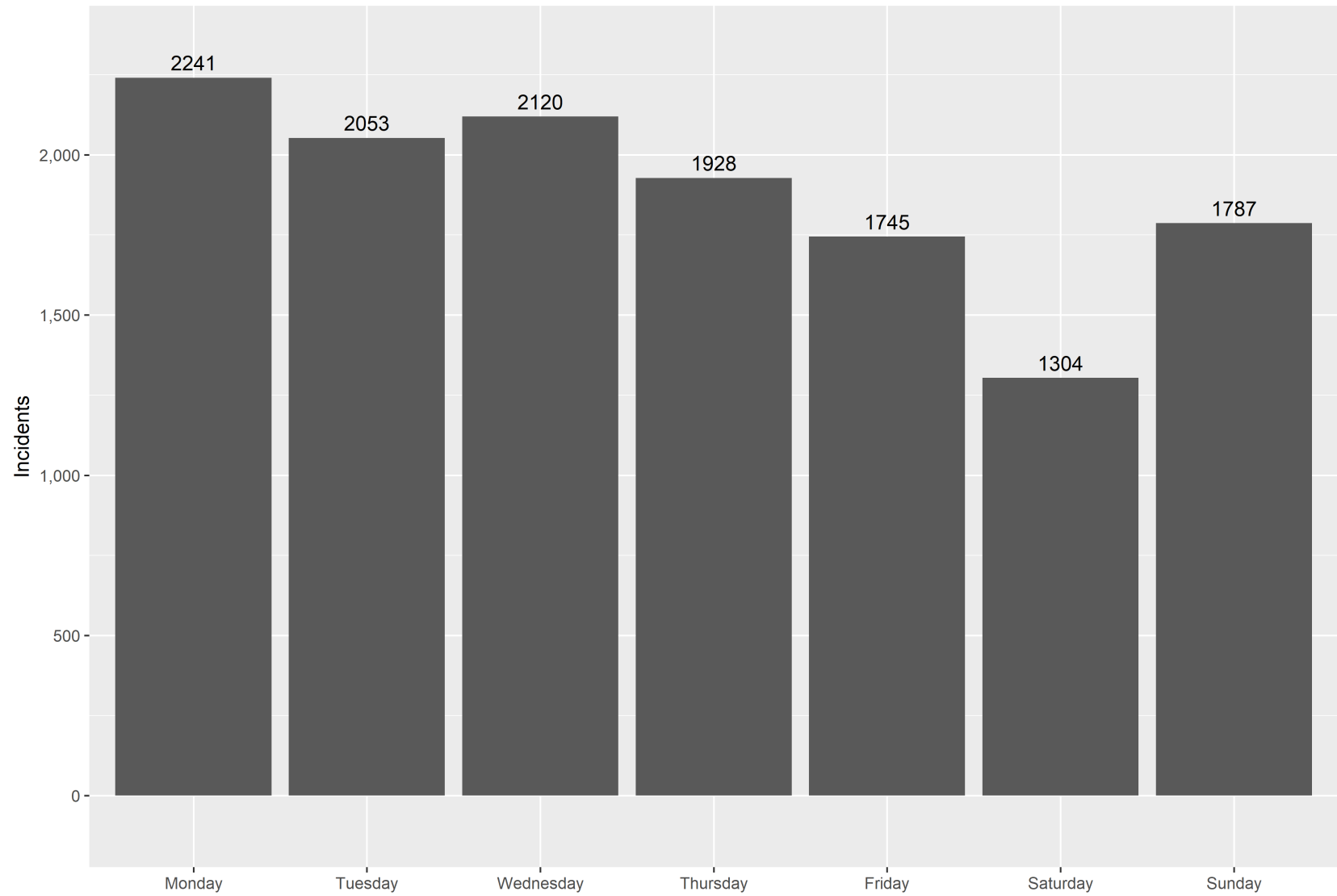
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Incident Types, 2017-2023

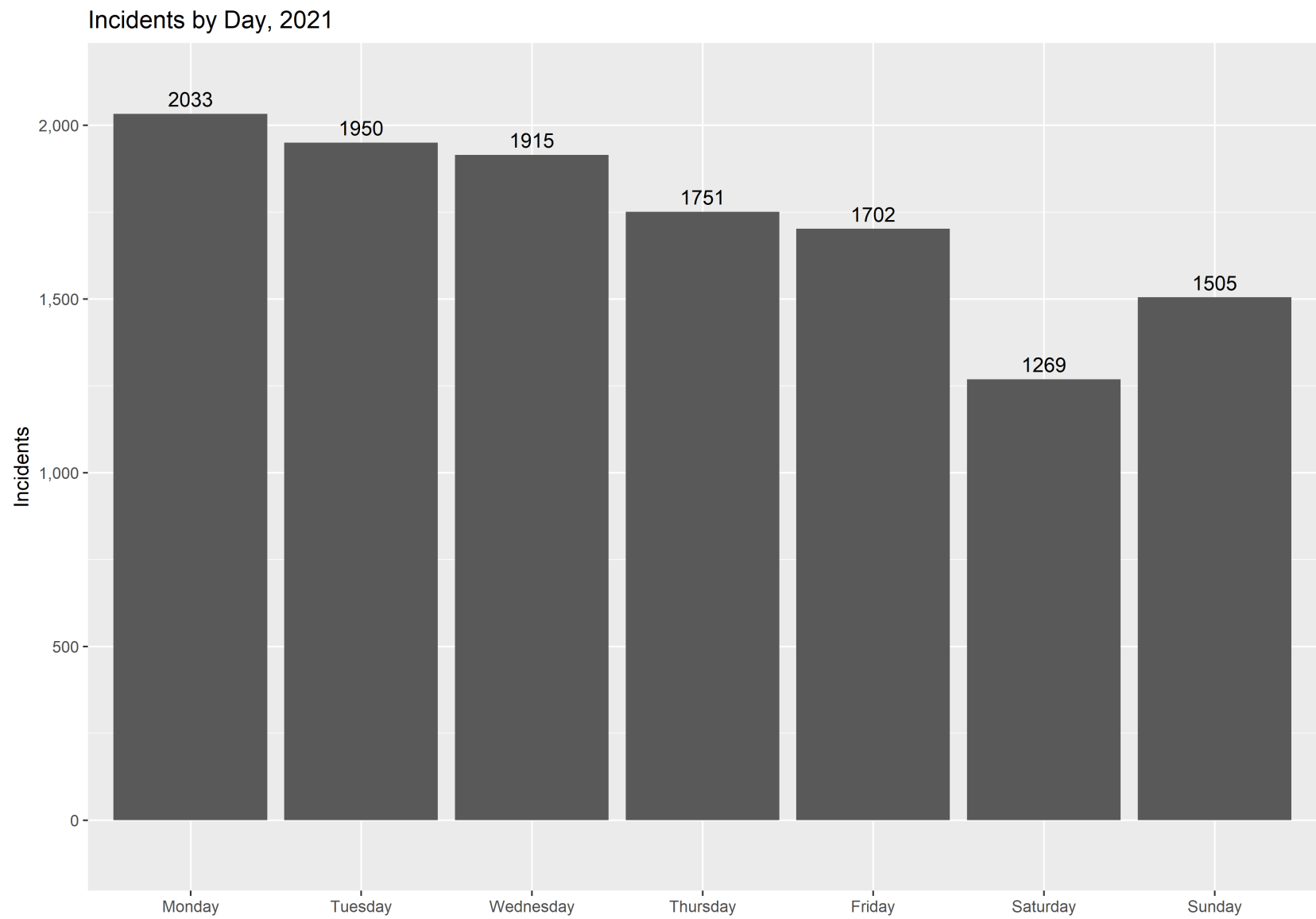


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Incidents by Day, 2023 YTD

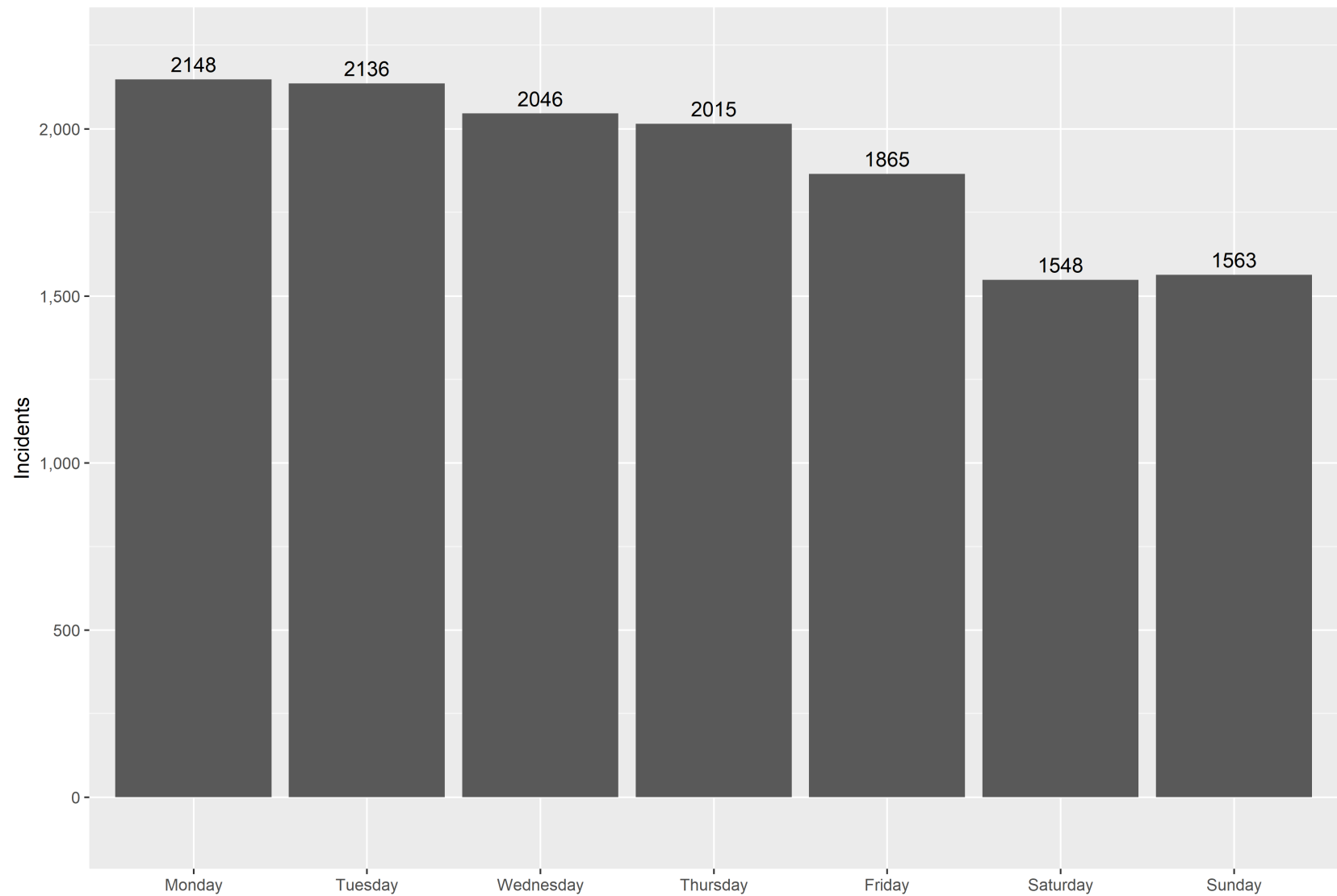


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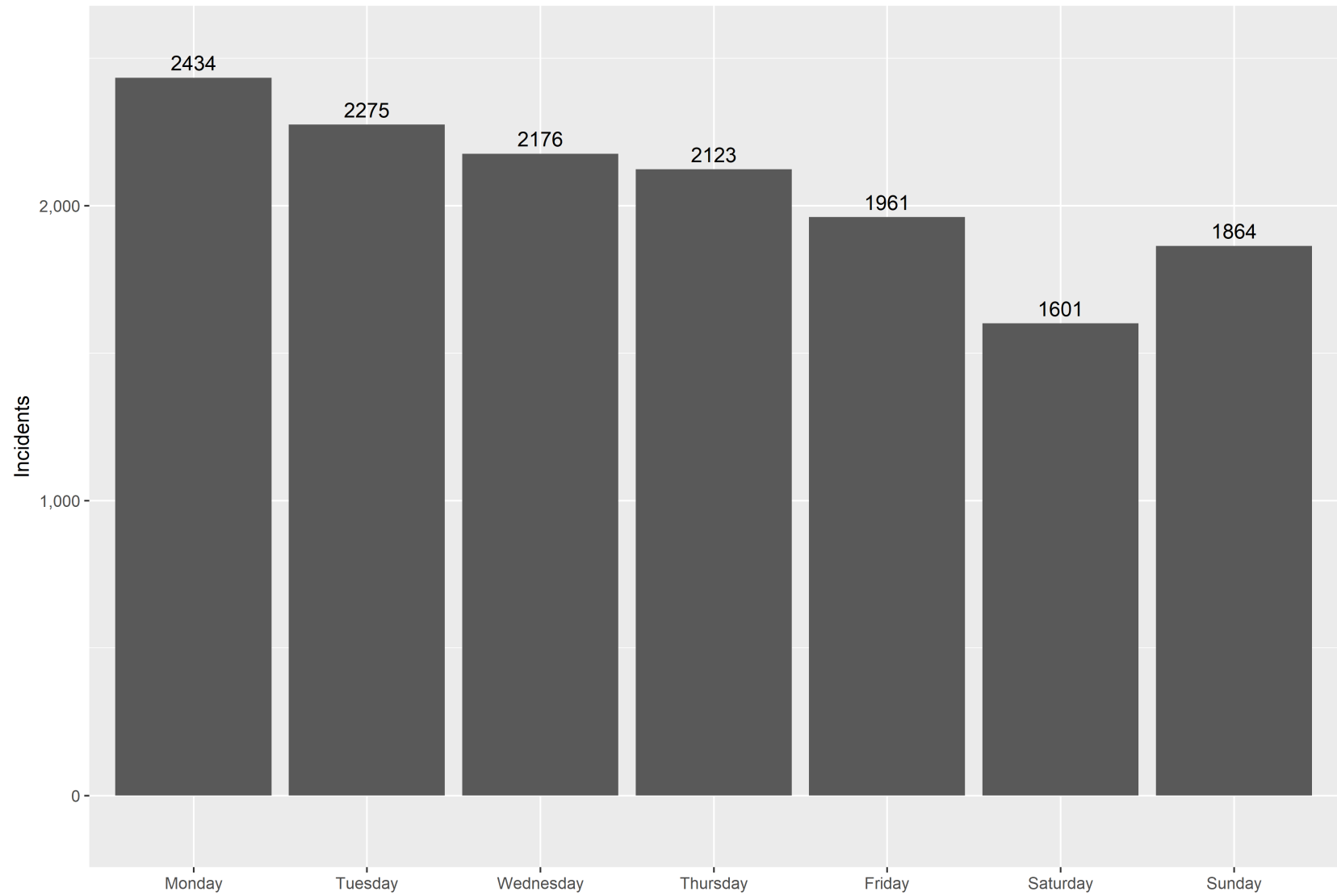
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Incidents by Day, 2020

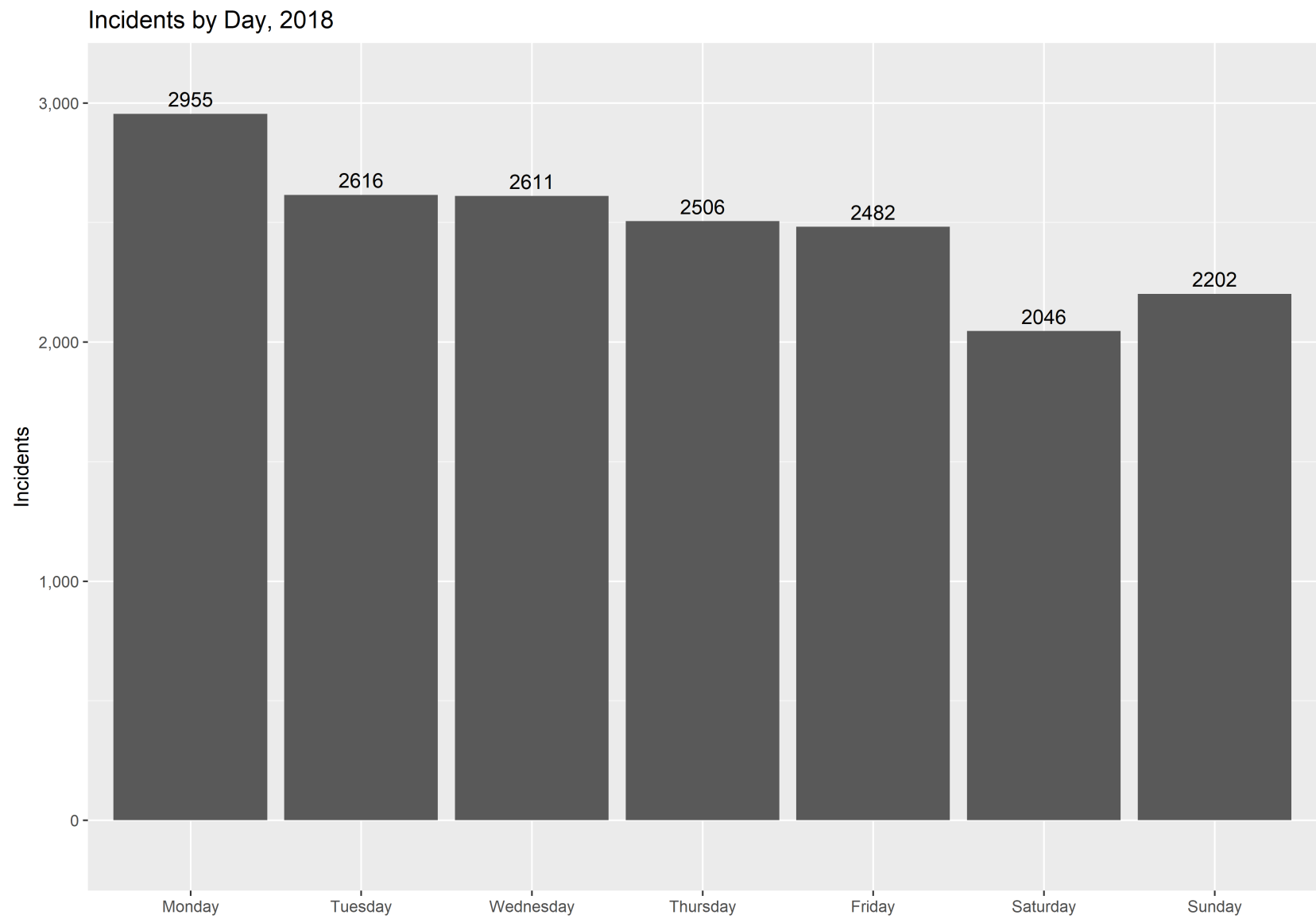


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Incidents by Day, 2019

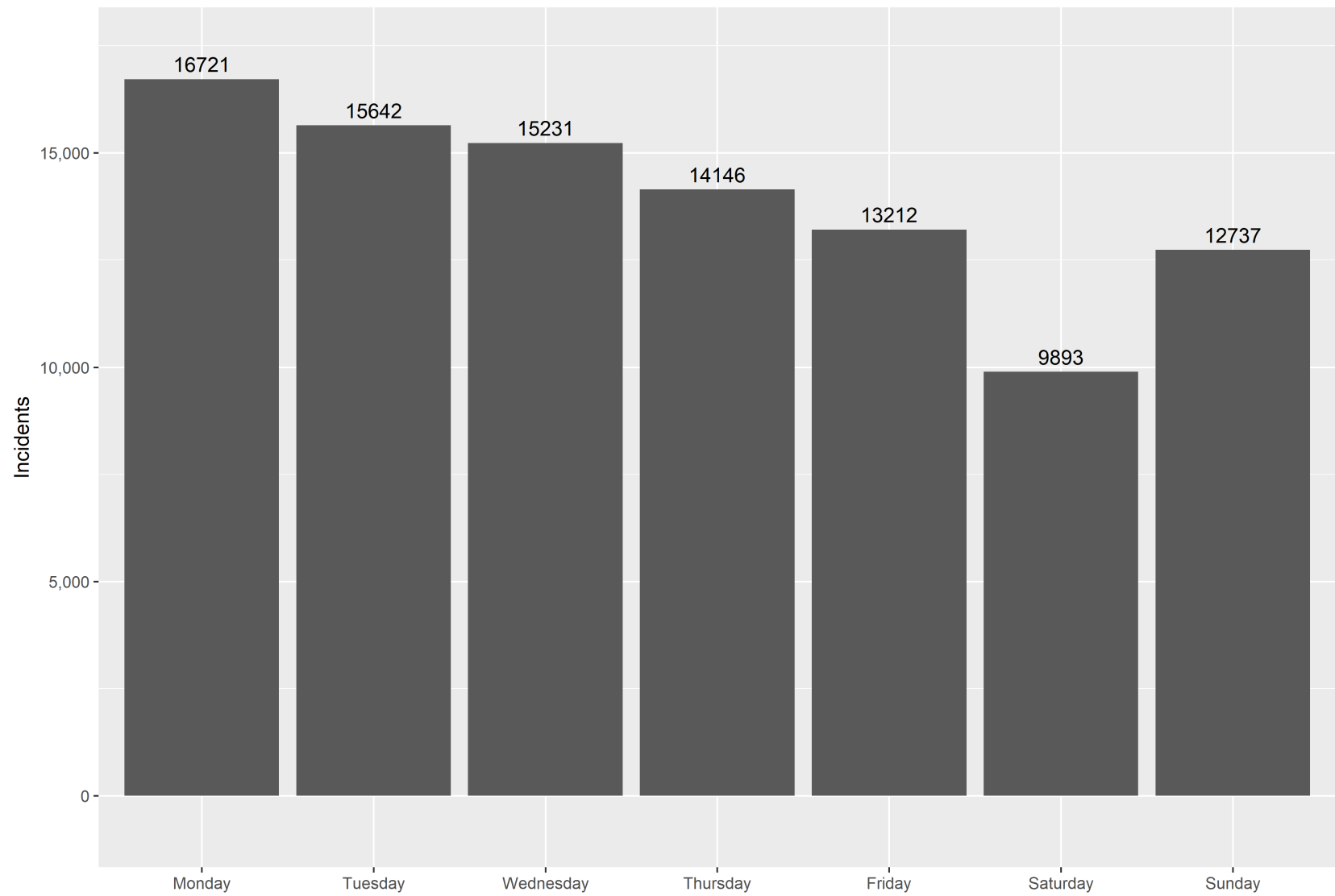


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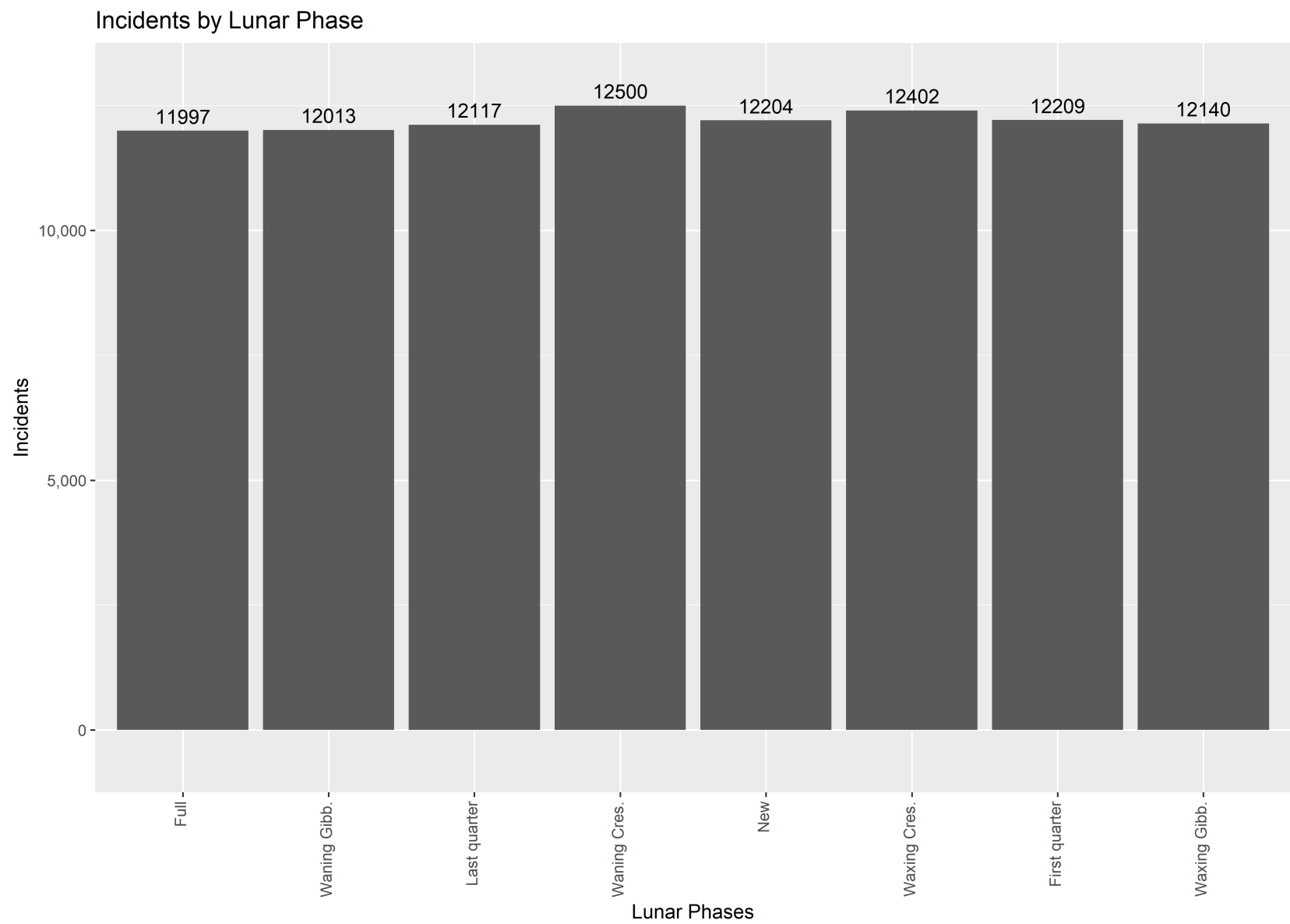


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Spokane Incidents by Day 2017 - 2023

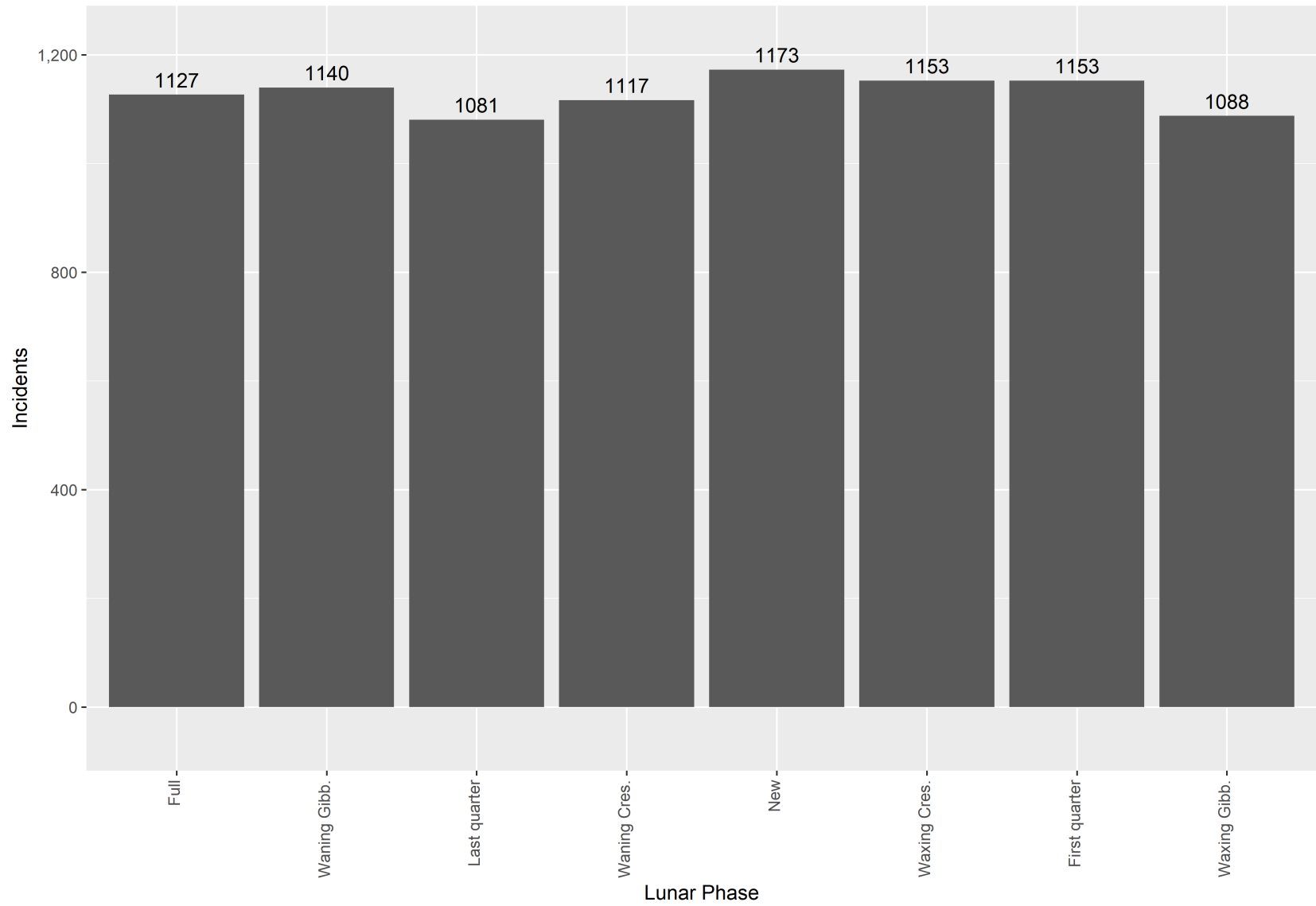


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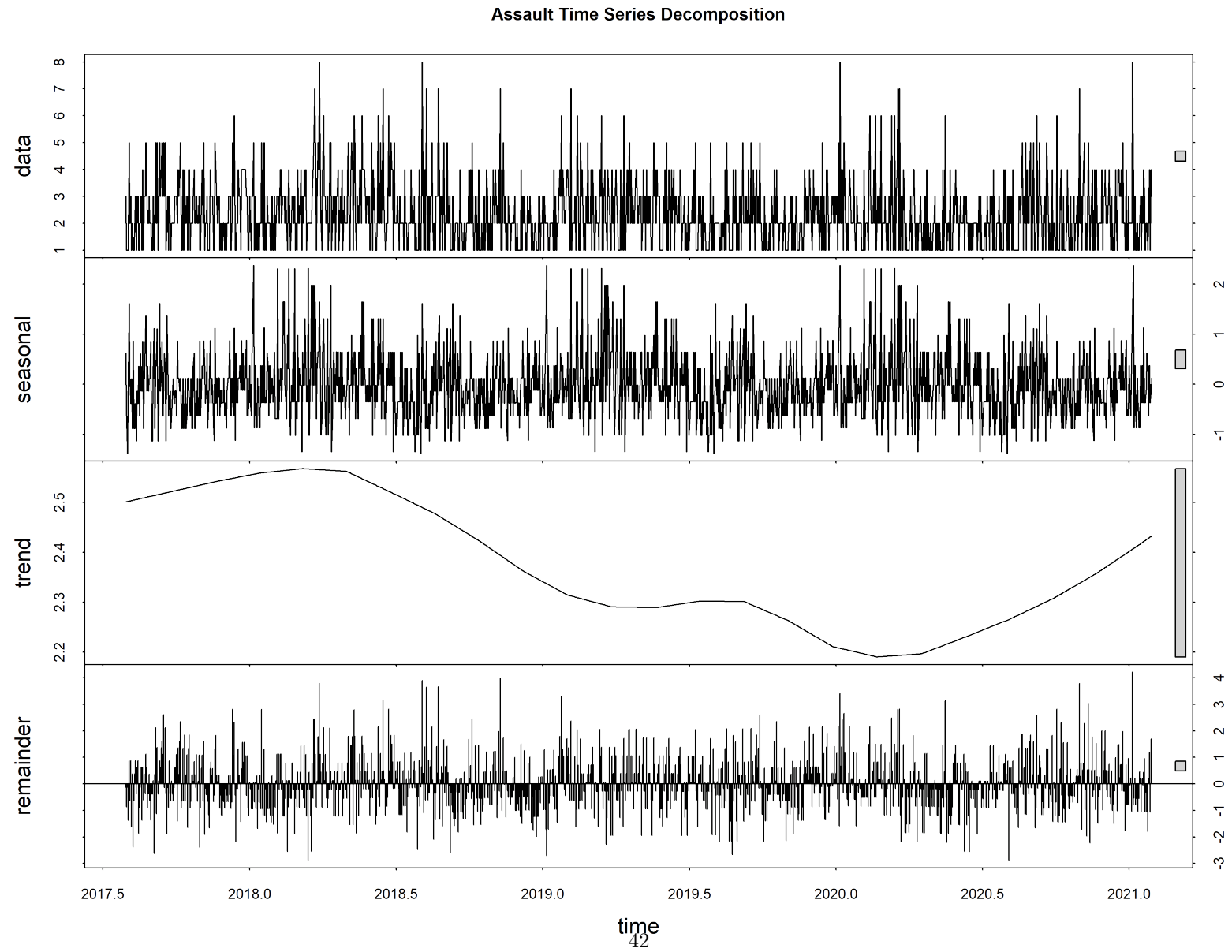
Violent Incidents by Lunar Phase



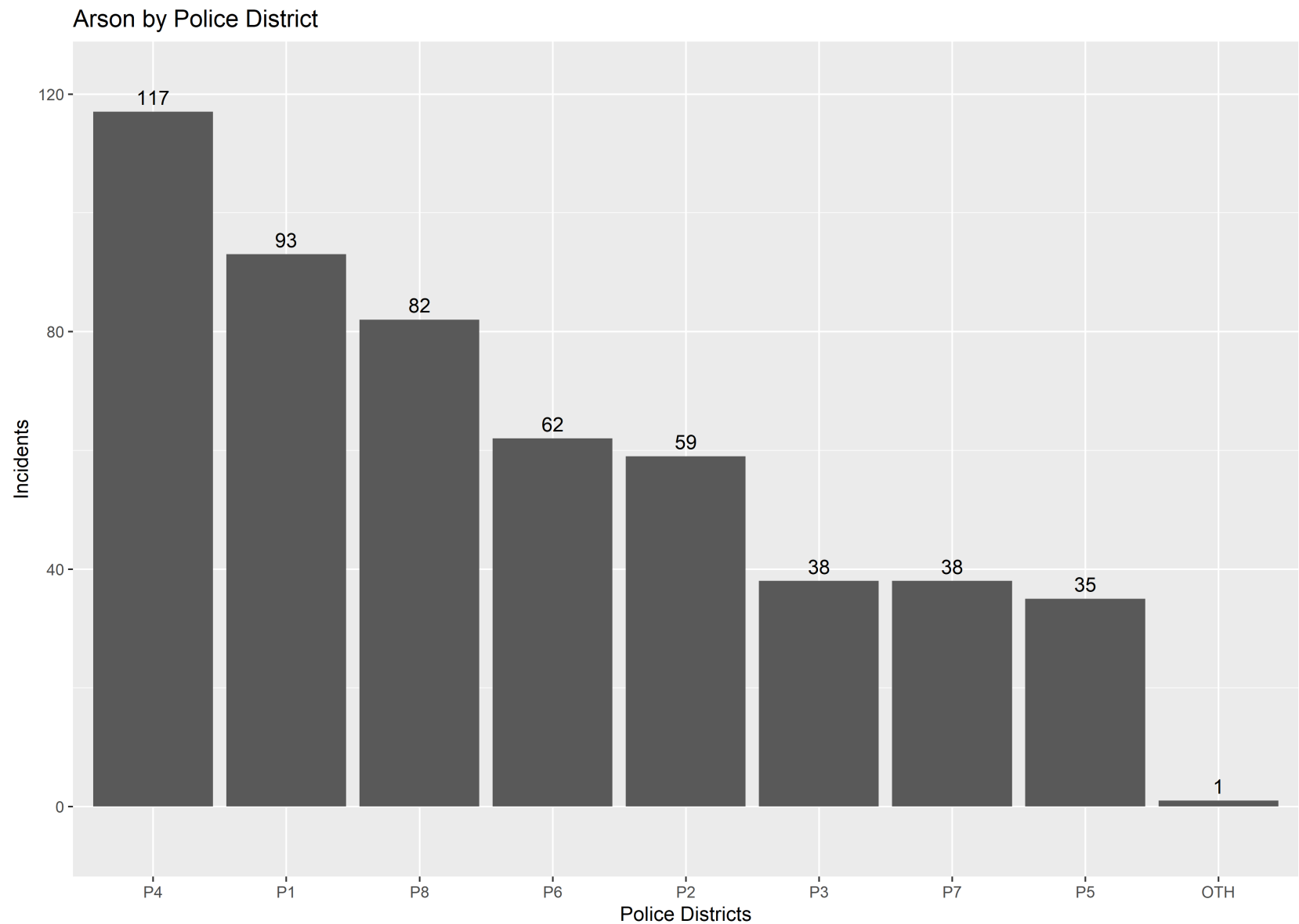
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4 Seasonal

4.1 Assault

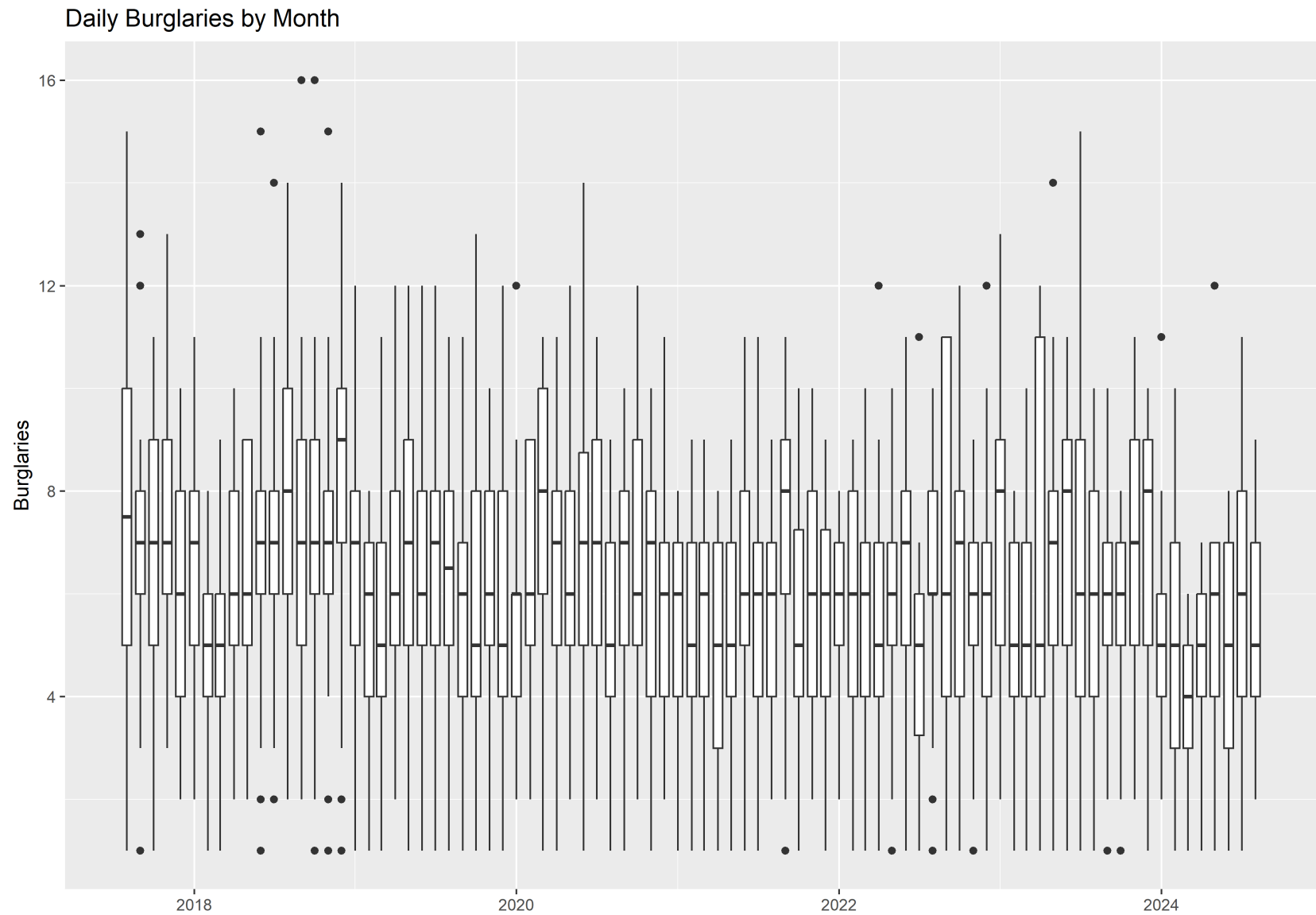


4.2 Arson

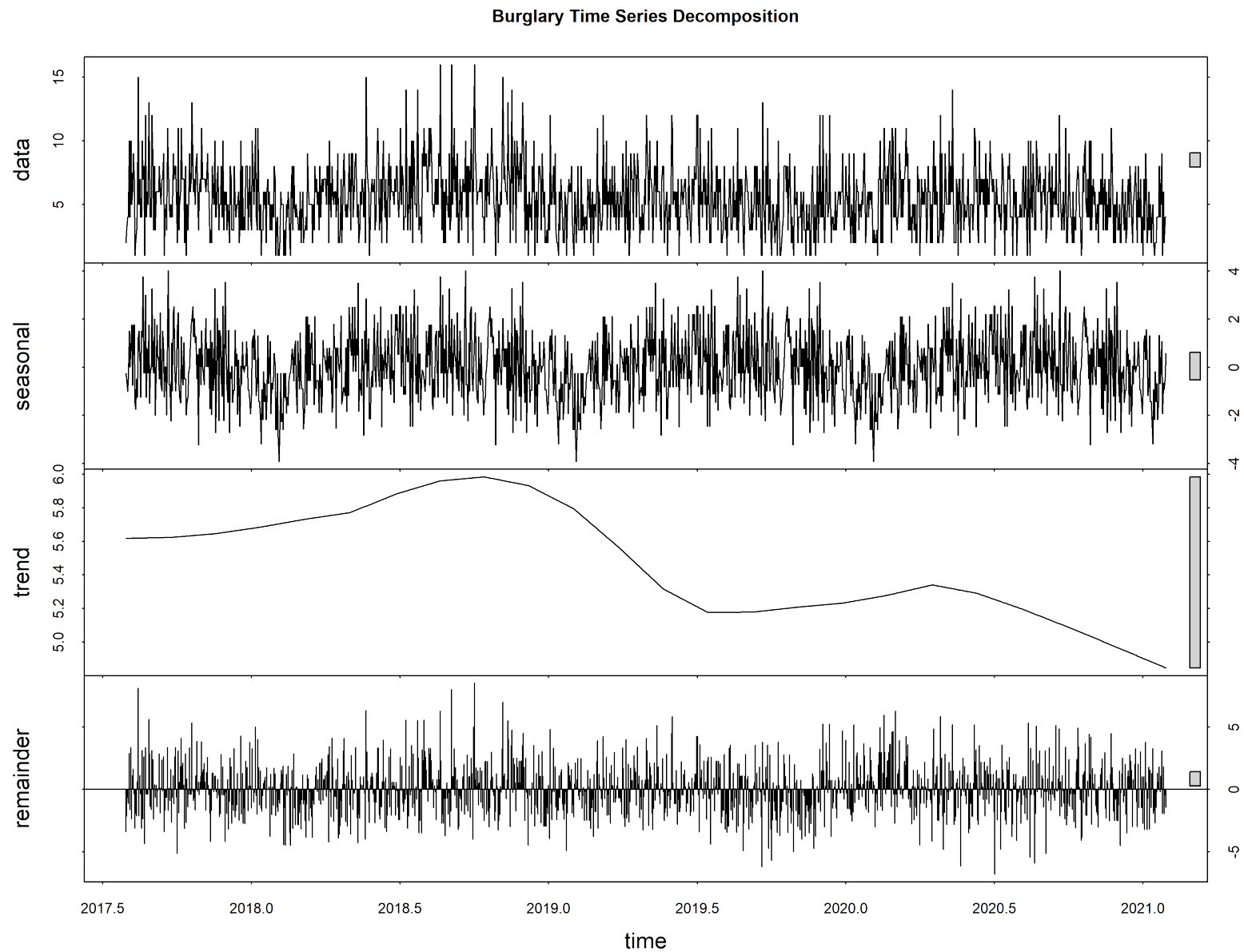


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4.3 Burglary

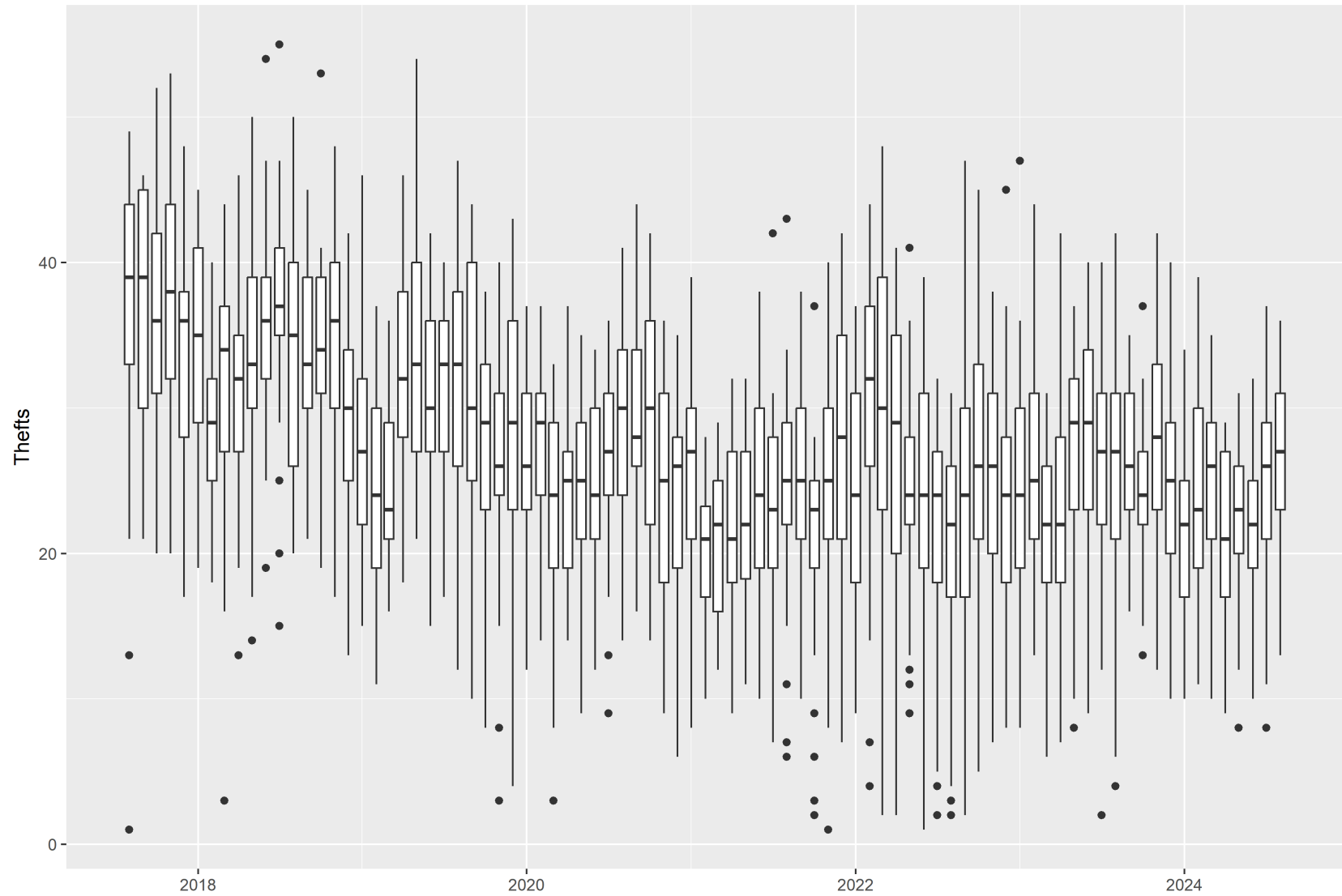


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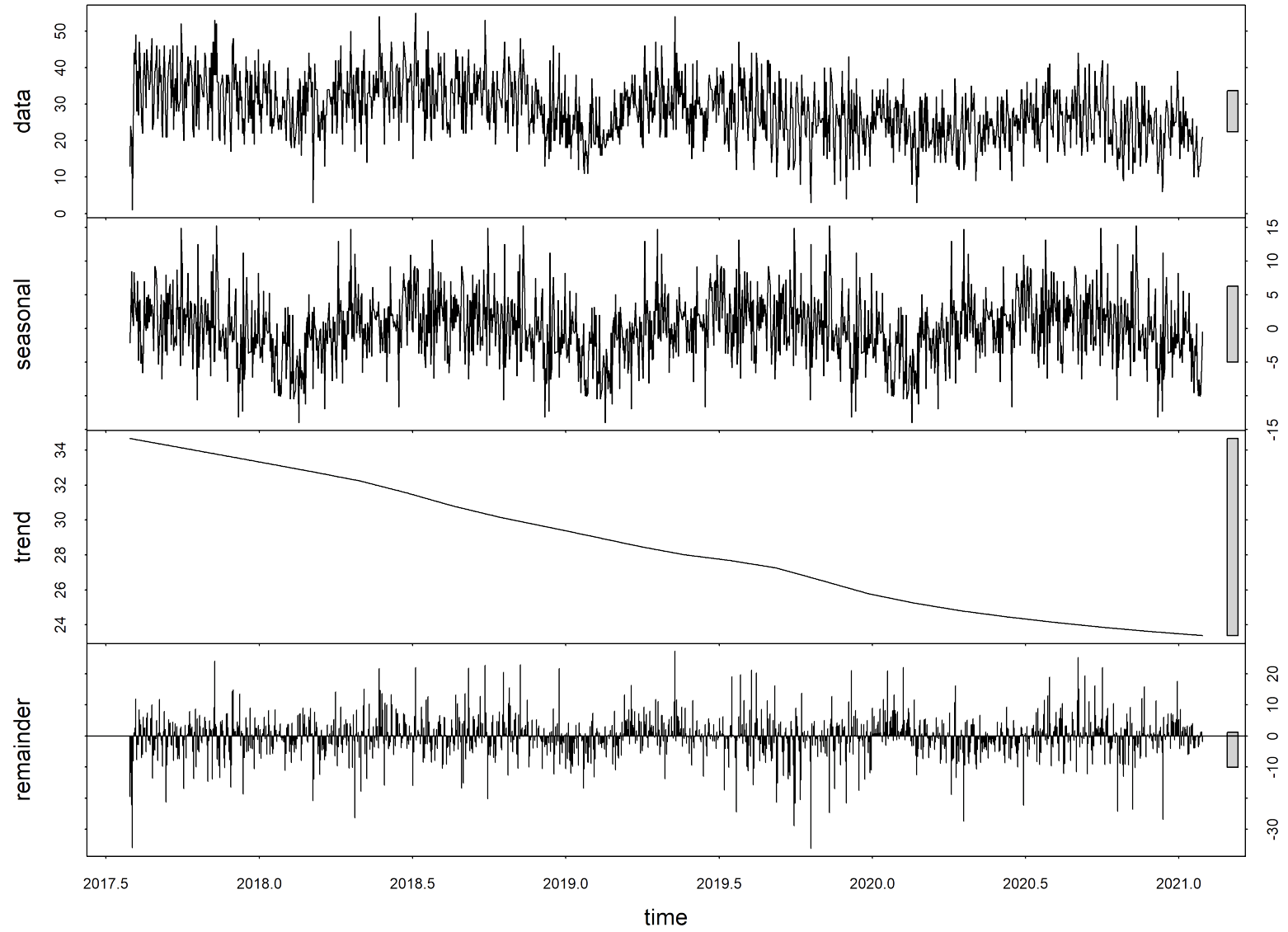
4.4 Theft

Daily Thefts by Month

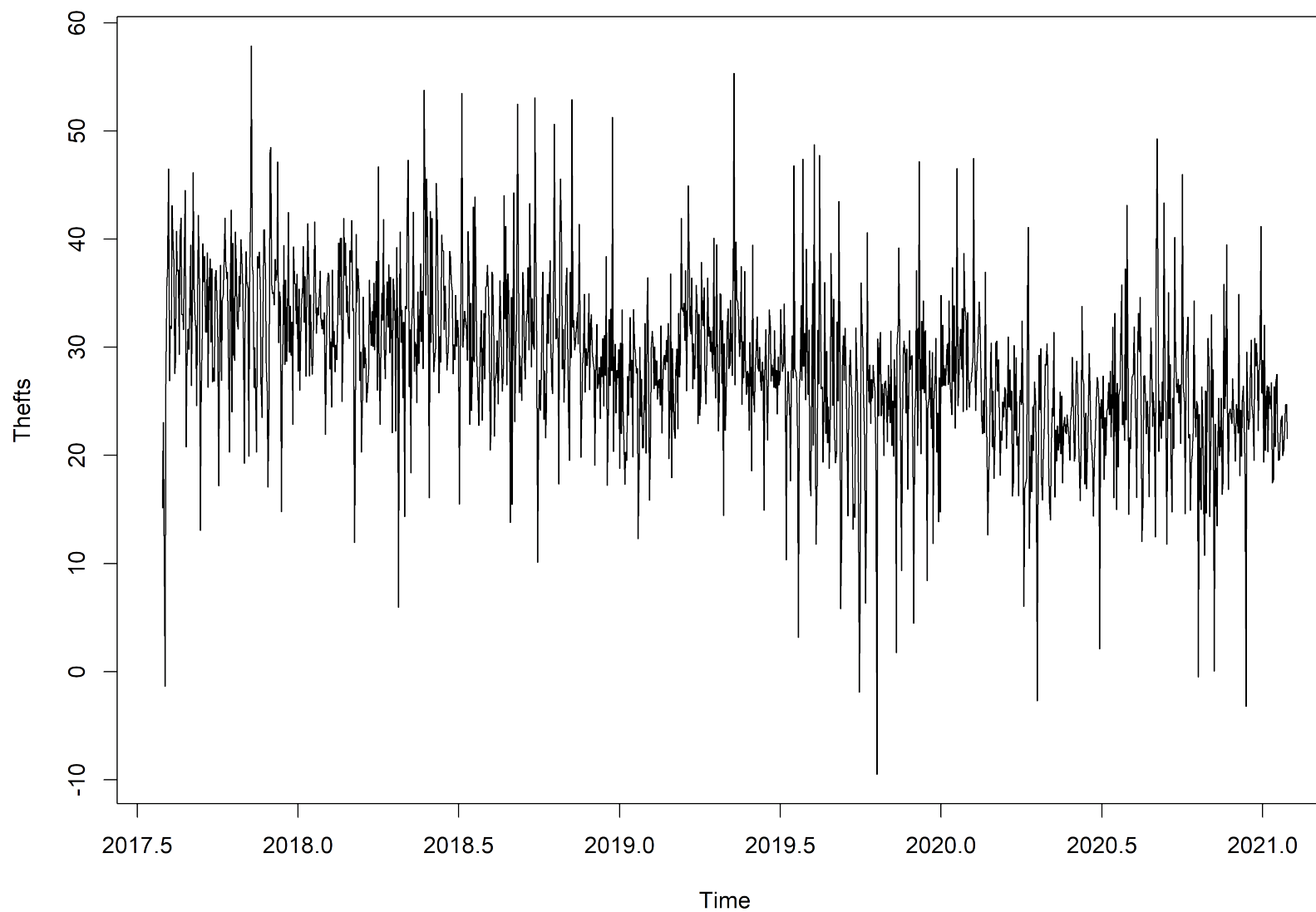


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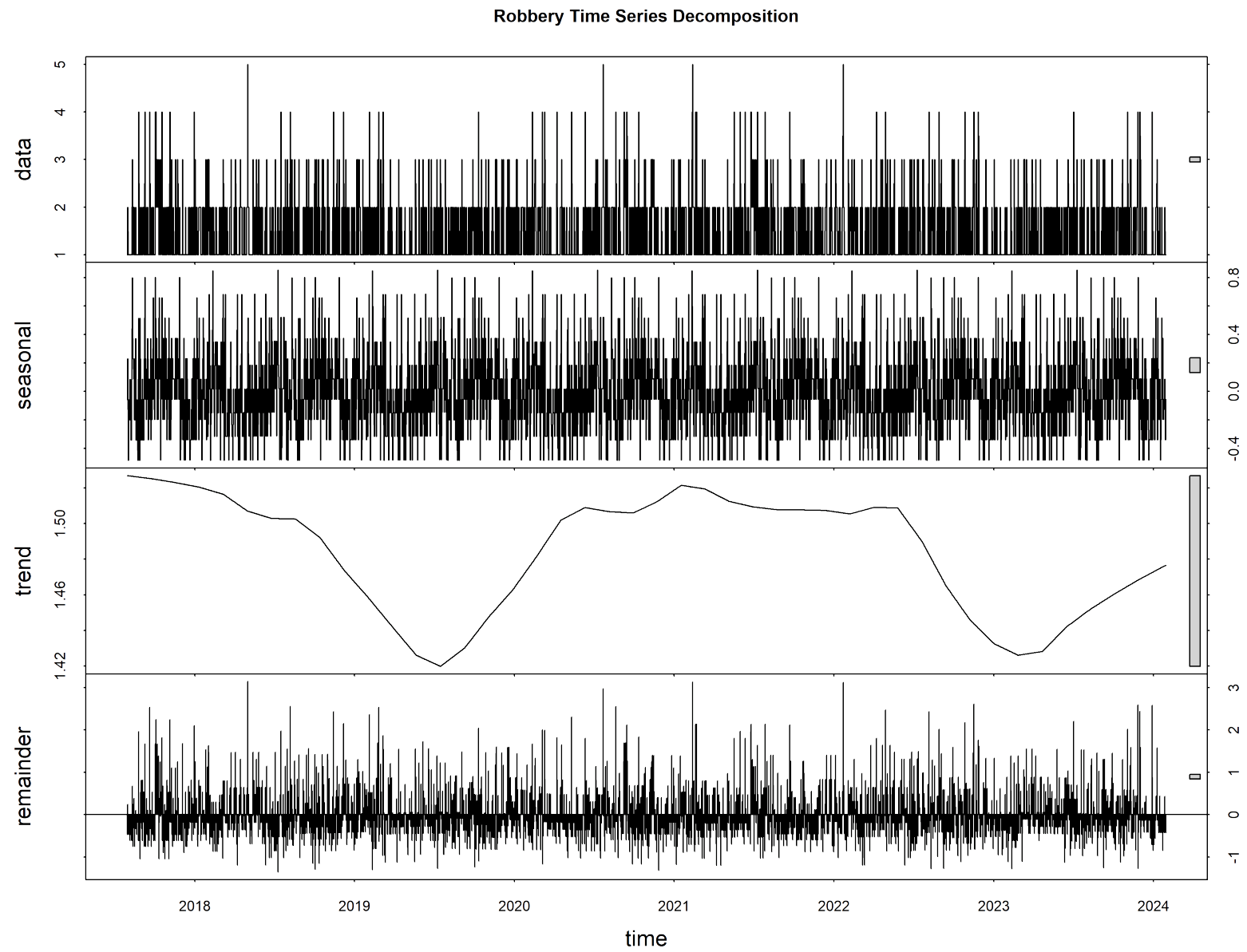
Theft Time Series Decomposition



Thefts over Time, Seasonally Adjusted

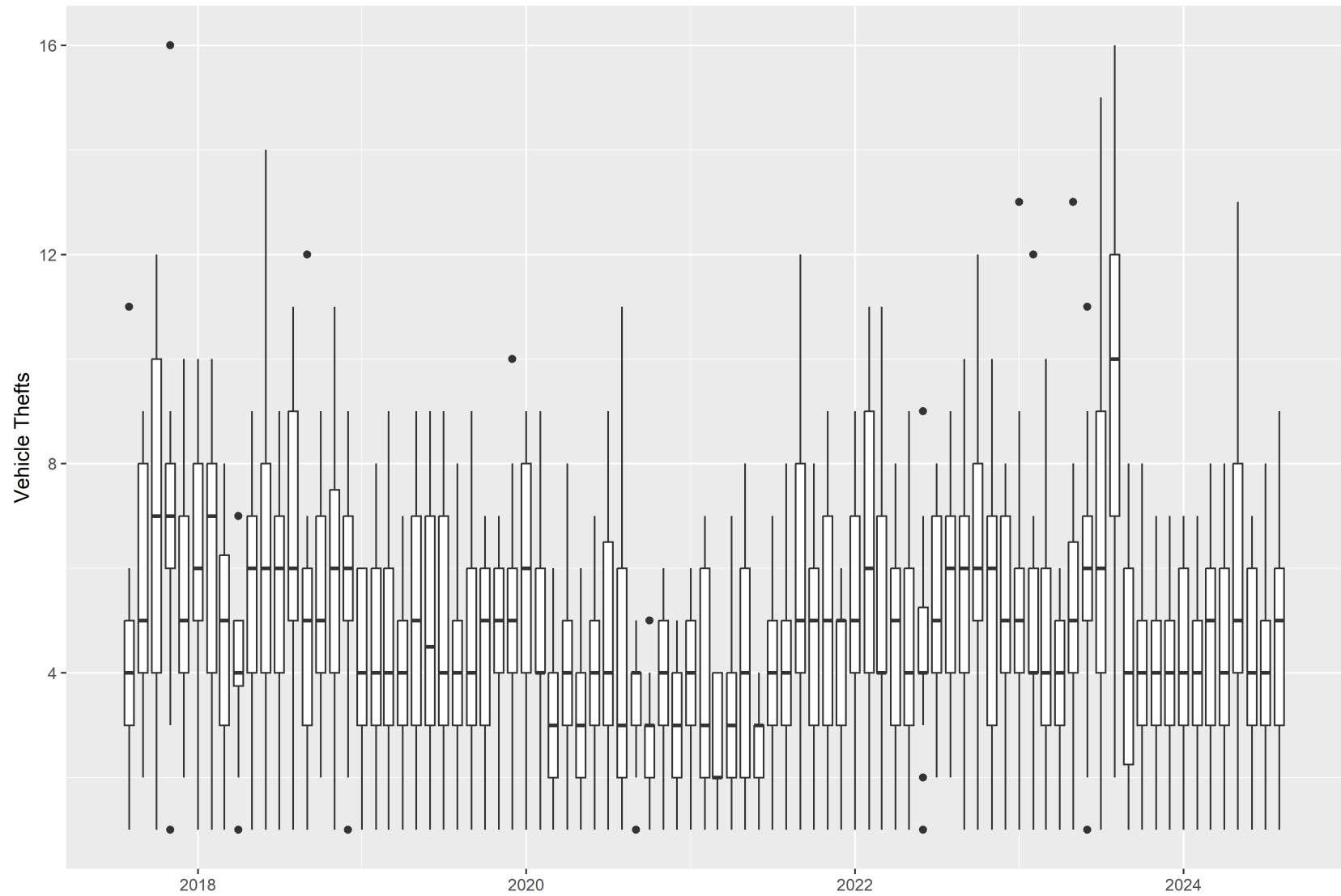


4.5 Robbery



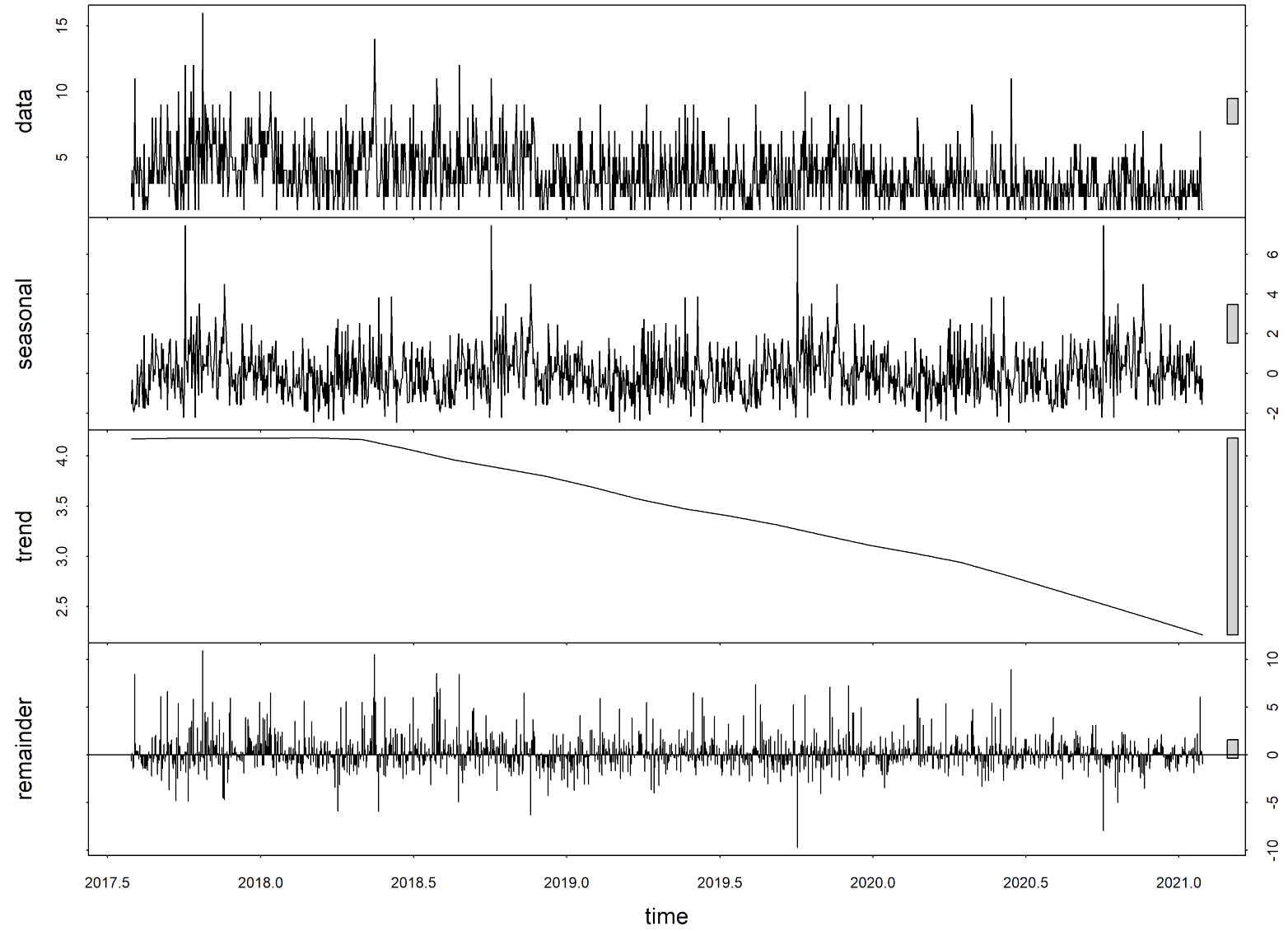
4.6 Motor Vehicle Theft

Daily Vehicle Thefts by Month



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Motor Vehicle Theft Time Series Decomposition



Vehicle Thefts over Time, Seasonally Adjusted

